

# Lab files

Thursday, July 10, 2025 1:02 PM

[http://github.com/MalwareCube/SOC101/blob/main/course\\_files/02\\_Network\\_Security.zip](http://github.com/MalwareCube/SOC101/blob/main/course_files/02_Network_Security.zip)

File password :

LCty2JmHQLB3uc9VxjeohENr

# Network Security Theory

Sunday, June 29, 2025 1:37 PM

This a refresh I you don't know or have networking basics => jermmy IT lab CCNA in 60 days on YouTube

## Network Layer Protocols

- **Internet Protocol (IP)**
  - IP addressing
  - IP routing
- **Transmission Control Protocol (TCP)**
  - Connection-oriented - reliable
  - Three-way handshake
  - Flow and congestion control
- **User Datagram Protocol (UDP)**
  - Connectionless
  - Lightweight, low-overhead - speed over reliability
  - VoIP, video streaming, audio streaming

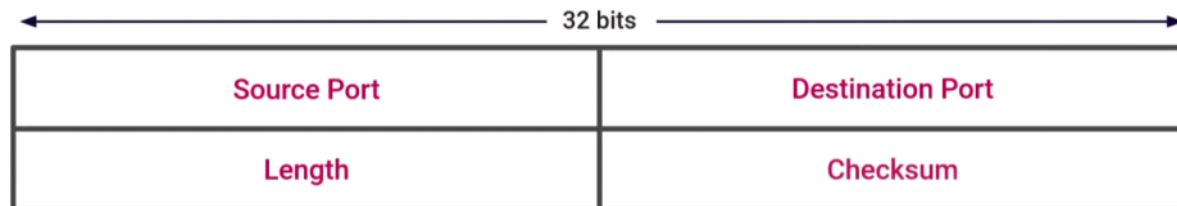


TCM

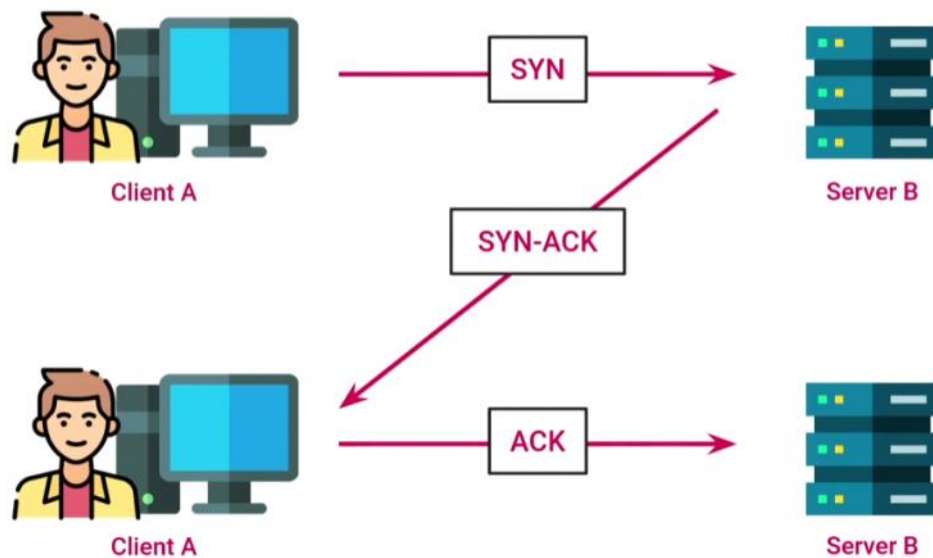
Internet Protocol Version 4 (IPv4) Header

|                     |               |                 |                 |                 |
|---------------------|---------------|-----------------|-----------------|-----------------|
| 32 bits             |               |                 |                 |                 |
| Version             | Header Length | Type of Service | Total Length    |                 |
| Identification      |               |                 | Flags           | Fragment Offset |
| Time To Live (TTL)  |               | Protocol        | Header Checksum |                 |
| Source Address      |               |                 |                 |                 |
| Destination Address |               |                 |                 |                 |
| Options             |               |                 | Padding         |                 |

### User Datagram Protocol (UDP) Header



## Three-Way Handshake

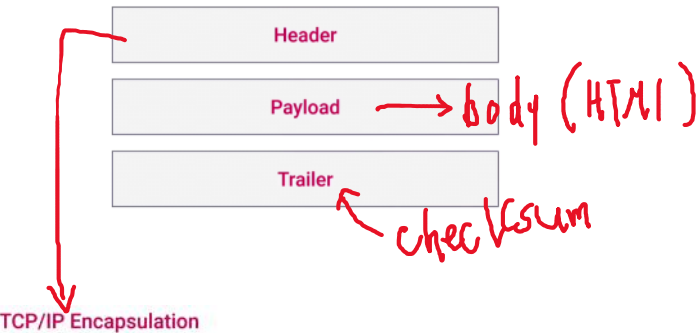


# Common Ports

- **21 - File Transfer Protocol (FTP)**
  - Transfer files between systems
- **22 - Secure Shell (SSH)**
  - Often used for a secure remote command-line interface
- **23 - Telnet**
  - Unencrypted remote command-line interface
- **25 - Simple Mail Transfer Protocol (SMTP)**
  - Sending emails
- **53 - Domain Name System (DNS)**
  - Translating domain names into IP addresses
- **80 - HyperText Transfer Protocol (HTTP)**
  - Data communication over the web
- **110 - Post Office Protocol 3 (POP3)**
  - Retrieving emails from a remote server
- **135 - Microsoft Remote Procedure Call (RPC)**
  - Microsoft inter-process communication
- **139 - NetBIOS**
  - Windows file and printer sharing
- **143 - Internet Message Access Protocol (IMAP)**
  - Retrieving emails from a remote server
- **389 - Lightweight Directory Access Protocol (LDAP)**
  - Centralized directory and authentication service
- **443 - HyperText Transfer Protocol (HTTPS)**
  - HTTP with encrypted communications
- **445 - Server Message Block (SMB)**
  - Network file and printer sharing
- **3389 - Remote Desktop Protocol (RDP)**
  - Graphical interface to connect with a system
- **8080 - HyperText Transfer Protocol (HTTP)**



### Packets



The PCAP is about seeing the routing info and the details

### Packet Capture (PCAP)

- Intercepting Packets**
  - Port mirroring (SPAN Ports) → Make a switch to copy traffic from port or more and sent to analysis
  - Inline network devices → Capture packets and analyze directly like the IPS or IDS
  - Network taps → Deceives connected to the network that can capture the traffic
- PCAP Files**
  - Binary file format
  - Header and raw payload
  - Tcpdump, Wireshark, Tshark, etc.
- Advantages**
  - Most detailed record
  - Includes packet payload
- Disadvantages**
  - Resource intensive
  - Requires significant storage space
  - Hard to scale

| Date       | Time  | Call Type | Number Dialed    | Duration | Cost    | Transcript                    |
|------------|-------|-----------|------------------|----------|---------|-------------------------------|
| 2024-05-01 | 14:35 | Local     | (555) 123-4567   | 10       | \$1.00  | ▶ "Hello. I'm just calli..."  |
| 2024-05-01 | 16:53 | Local     | (555) 987-6543   | 5        | \$0.50  | ▶ "Hey Bill, do you thi..."   |
| 2024-05-02 | 09:12 | Long Dist | (800) 555-1234   | 20       | \$5.00  | ▶ "Is this the Krusty K..."   |
| 2024-05-02 | 18:06 | Intl      | +44 20 7946 0958 | 15       | \$10.00 | ▶ "Hi there. You hav..."      |
| 2024-05-03 | 20:45 | Local     | (555) 555-5555   | 30       | \$3.00  | ▶ "Congratulations! Y..."     |
| 2024-05-04 | 07:27 | Long Dist | (888) 555-6789   | 25       | \$6.25  | ▶ "You're late for work!"     |
| 2024-05-05 | 22:02 | Intl      | +81 3-1234-5678  | 10       | \$7.50  | ▶ "Why do you keep c..."      |
| 2024-05-05 | 15:25 | Local     | (555) 222-3333   | 40       | \$4.00  | ▶ "Hi there. My name i..."    |
| 2024-05-06 | 19:12 | Local     | (555) 444-4444   | 12       | \$1.20  | ▶ "Are you free this w..."    |
| 2024-05-06 | 21:42 | Long Dist | (900) 555-7890   | 18       | \$4.50  | ▶ "Hi Bill. It's Mary. Do..." |

Flow records is about normal and routing information from routers

### Flow Records

- Aggregated metadata**
- 5-Tuple**
  - Source IP
  - Source Port
  - Destination IP
  - Destination Port
  - Transport Protocol
- Advantages**
  - Efficient bandwidth and storage requirements
  - High level pattern and anomaly detection
  - Easier to scale
- Disadvantages**
  - Lacks payload detail
  - Doesn't give you the whole picture

| Date       | Time  | Call Type | Number Dialed    | Duration (min) | Cost    |
|------------|-------|-----------|------------------|----------------|---------|
| 2024-05-01 | 14:35 | Local     | (555) 123-4567   | 10             | \$1.00  |
| 2024-05-01 | 16:53 | Local     | (555) 987-6543   | 5              | \$0.50  |
| 2024-05-02 | 09:12 | Long Dist | (800) 555-1234   | 20             | \$5.00  |
| 2024-05-02 | 18:06 | Intl      | +44 20 7946 0958 | 15             | \$10.00 |
| 2024-05-03 | 20:45 | Local     | (555) 555-5555   | 30             | \$3.00  |
| 2024-05-04 | 07:27 | Long Dist | (888) 555-6789   | 25             | \$6.25  |
| 2024-05-05 | 22:02 | Intl      | +81 3-1234-5678  | 10             | \$7.50  |
| 2024-05-05 | 15:25 | Local     | (555) 222-3333   | 40             | \$4.00  |
| 2024-05-06 | 19:12 | Local     | (555) 444-4444   | 12             | \$1.20  |
| 2024-05-06 | 21:42 | Long Dist | (900) 555-7890   | 18             | \$4.50  |

## Introduction to tcpdump

Monday, July 7, 2025 12:45 PM

Is a command line packet capturing tool

On windows we have windump

```
$ sudo apt-get install tcpdump
```

- In this tool we can do packet capture as well as we can make some scripts to capture as an event happens
- It captures packets directly from the network interface and it connects to the low-level OS

This as the filter or choosing interface options

```
sudo tcpdump [options] [expression]
```

This where we choose the thing the option will work on. As IP or network interface

To show the network interfaces you have :

```
m8a8a8y@m8a8a8y-VirtualBox: $ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd00::a00:27ff:fea4:86ed prefixlen 64 scopeid 0x0<global>
    inet6 fd00::1c0:350a:fc20:7ea prefixlen 64 scopeid 0x0<global>
    inet6 fe00::a00:27ff:fea4:86ed prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:a4:86:ed txqueuelen 1000 (Ethernet)
    RX packets 268 bytes 323948 (323.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 178 bytes 22396 (22.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 44 bytes 5484 (5.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 44 bytes 5484 (5.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Or

```
m8a8a8y@m8a8a8y-VirtualBox: $ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:a4:86:ed brd ff:ff:ff:ff:ff:ff
```

This command will show the interfaces that tcpdump can use:

```
m8a8a8y@m8a8a8y-VirtualBox: $ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
```

Using this command you can select the interface either the number or with the name (lo for 3)

```
m8a8a8y@m8a8a8y-VirtualBox: $ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
m8a8a8y@m8a8a8y-VirtualBox: $ sudo tcpdump -i 3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:10:00.854634 IP localhost.51900 > _localdnsstub.domain: 9629+ AAAA? api.snapcraft.io. (34)
13:10:00.854791 IP localhost.57218 > _localdnsstub.domain: 56556+ A? api.snapcraft.io. (34)
```

```

m8a8a8y@m8a8a8y-VirtualBox:~$ tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i 3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:10:00.854634 IP localhost.51900 > _localdnsstub.domain: 9629+ AAAA? api.snapcraft.io. (34)
13:10:00.854791 IP localhost.57218 > _localdnsstub.domain: 56556+ A? api.snapcraft.io. (34)
13:10:00.870621 IP _localdnsstub.domain > localhost.51900: 9629 4/3/0 AAAA 2620:2d:4000:1010::2e6, AAAA 2620:2d:4000:1010::42, AAAA 2620:2d:4000:1010::117 (213)
13:10:00.872820 IP _localdnsstub.domain > localhost.57218: 56556 4/3/0 A 185.125.188.58, A 185.125.188.57, A 185.125.188.54, A 185.125.188.59 (165)
13:10:00.893136 IP localhost.56598 > _localdnsstub.domain: 64528+ [1au] PTR? 53.0.0.127.in-addr.arpa. (52)
13:10:00.893661 IP _localdnsstub.domain > localhost.56598: 64528*$ 1/0/1 PTR _localdnsstub. (79)

```

Here we used net cat to make a listening section and on a port and another terminal listen contact that port and tcpdump see the packets

```

m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i 3
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:16:54.918639 IP localhost.42700 > _localdnsstub.domain: 32100+ [1au] PTR? 0.0.0.0.in-addr.arpa. (49)
13:16:54.918851 IP _localdnsstub.domain > localhost.42700: 32100 NXDomain*$ 0/0/1 (49)
13:16:55.000261 IP _localdnsstub.domain > _localdnsstub.domain: 43009+ [1au] PTR? 53.0.0.127.in-addr.arpa. (52)
13:16:55.000552 IP _localdnsstub.domain > localhost.48624: 43009*$ 1/0/1 PTR _localdnsstub. (79)
13:16:57.277674 IP localhost.53796 > localhost.4321: Flags [S], seq 2930519626, win 65495, options [mss 65495,sackOK,TS val 26669280 ecr 0,nop,wscale 7], length 0
13:16:57.277691 IP localhost.4321 > localhost.53796: Flags [S.], seq 686879756, ack 2930519627, win 65483, options [mss 65495,sackOK,TS val 26669280 ecr 26669280,nop,wscale 7], length 0
13:16:57.277705 IP localhost.53796 > localhost.4321: Flags [.] , ack 1, win 512, options [nop,nop,TS val 26669280 ecr 26669280], length 0
13:17:03.461624 IP localhost.mdns > mdns.mcast.net.mdns: 0 [ZQ] PTR (QM)? _lpp._tcp.local. PTR (QM)? _lpps._tcp.local. (45)
13:17:03.565402 IP localhost.37002 > _localdnsstub.domain: 65439+ [1au] PTR? 251.0.0.224.in-addr.arpa. (53)
13:17:03.582309 IP _localdnsstub.domain > localhost.37002: 65439 1/0/1 PTR mdns.mcast.net. (81)
13:17:09.052765 IP localhost.4321 > localhost.53796: Flags [P.], seq 1:7, ack 1, win 512, options [nop,nop,TS val 26681055 ecr 26669280] length 6
13:17:09.052852 IP localhost.53796 > localhost.4321: Flags [.] , ack 7, win 512, options [nop,nop,TS val 26681055 ecr 26681055], length 0

```

```

m8a8a8y@m8a8a8y-VirtualBox:~$ nc -lvp 4321
Listening on 0.0.0.0 4321
connection received on localhost 53796
hello

```

```

m8a8a8y@m8a8a8y-VirtualBox:~$ nc 127.0.0.1 4321
hello

```

when the connection was received tcpdump so shows Sin [S] Sin-Ack [S.]  
Ack [.]

when we sent hello we saw the push-Ack [P.] Ack [.]

If you saw the blue circle this the length of the packet this means there is somethings sent and next we will see how to view it

We restarted the listening on tcpdump but we added the -X ( will print out the hex of message and the ASCII of it to be read format  
(if we used -A we are going to see only ASCII no hex)

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i lo -X
[sudo] password for m8a8a8y:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
13:36:03.069361 IP localhost.4321 > localhost.53796: Flags [P.], seq 686879763:686879769, ack 2938519627, win 512, options [nop,nop,TS val 27815071 ecr 26681055], length 6
    0x0000: 4500 003a 806d 4000 4006 bc4e 7f00 0001  E...m0.0..N....
    0x0010: 7f00 0001 10e1 d224 28f0 f413 aeac 2e4b  .....$(.....K
    0x0020: 8018 0200 fe2e 0000 0101 080a 01a8 6c9f  .....l.
    0x0030: 0197 1edf 6865 6c6c 6f0a                ..hello.
13:36:03.069583 IP localhost.53796 > localhost.4321: Flags [.], ack 0, win 512, options [nop,nop,TS val 27815071 ecr 27815071], length 0
    0x0000: 4500 0034 c56c 4000 4006 7755 7f00 0001  E..4.l0.0.wU....
    0x0010: 7f00 0001 d224 10e1 aeac 2e4b 28f0 f419  .....$(.....K(
    0x0020: 8010 0200 fe28 0000 0101 080a 01a8 6c9f  .....(.....l.
    0x0030: 01a8 6c9f                                ..l.
```

m8a8a8y@m8a8a8y-VirtualBox:~\$ nc -lvp 4321  
Listening on 0.0.0.0 4321  
Connection received on localhost 53796  
hello  
hello

m8a8a8y@m8a8a8y-VirtualBox:~\$ nc 127.0.0.1 4321  
hello  
hello  
[]



## tcpdump: Capturing Network Traffic

Tuesday, July 8, 2025 9:23 AM

- By default tcpdump translate ip addresses into there name and port name to remove it add -n

```
9277, ack 799, win 31970, length 506
20:32:52.564450 IP SOC101-ubuntu.57396 > 190.92.188.190.static.a2webhosting.com.https:
  win 0, length 0
^C
180 packets captured
180 packets received by filter
0 packets dropped by kernel
tcm@SOC101-ubuntu:~$ sudo tcpdump -i enp0s3 -n
```

- Host : will only filter if the host name is same as give here its tesla.com

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i enp0s3 -n host tesla.com
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:32:45.227069 IP 10.0.2.15.57606 > 2.18.51.207.443: Flags [S], seq 3395514882, win 64240, options [mss 1460,sackOK,TS val 810821854 ec
r 0,nop,wscale 7], length 0
09:32:45.245566 IP 10.0.2.15.57610 > 2.18.51.207.443: Flags [S], seq 3913323823, win 64240, options [mss 1460,sackOK,TS val 810821873 ec
r 0,nop,wscale 7], length 0
09:32:45.443614 IP 2.18.51.207.443 > 10.0.2.15.57606: Flags [S.], seq 47168001, ack 3395514883, win 65535, options [mss 1460], length 0
09:32:45.443718 IP 10.0.2.15.57606 > 2.18.51.207.443: Flags [.], ack 1, win 64240, length 0
09:32:45.444918 IP 10.0.2.15.57606 > 2.18.51.207.443: Flags [P.], seq 1:1895, ack 1, win 64240, length 1894
09:32:45.446076 IP 2.18.51.207.443 > 10.0.2.15.57606: Flags [.], ack 1461, win 65535, length 0
09:32:45.446078 IP 2.18.51.207.443 > 10.0.2.15.57606: Flags [.], ack 1895, win 65535, length 0
09:32:45.463602 IP 2.18.51.207.443 > 10.0.2.15.57610: Flags [S.], seq 47232001, ack 3913323824, win 65535, options [mss 1460], length 0
09:32:45.463647 IP 10.0.2.15.57610 > 2.18.51.207.443: Flags [.], ack 1, win 64240, length 0
09:32:45.465199 IP 10.0.2.15.57610 > 2.18.51.207.443: Flags [P.], seq 1:1895, ack 1, win 64240, length 1894
09:32:45.466154 IP 2.18.51.207.443 > 10.0.2.15.57610: Flags [.], ack 1461, win 65535, length 0
09:32:45.466155 IP 2.18.51.207.443 > 10.0.2.15.57610: Flags [.], ack 1895, win 65535, length 0
```

- Src & dst : are used to filter on source ip or destination ip

```
$ sudo tcpdump -i enp0s3 -n dst 142.251.33.174
```

- Net : is used to filter on specific subnet

```
$ sudo tcpdump -i enp0s3 -n net 192.168.1.0/24
```

```
$ sudo tcpdump -i enp0s3 -n src net 192.168.1.0/24
```

```
$ sudo tcpdump -i enp0s3 -n dst net 192.168.1.0/24
```

- Port will filter on a specific port

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -i enp0s3 -n port 443
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
09:40:32.390241 IP 10.0.2.15.42688 > 216.58.211.206.443: Flags [S], seq 3815115952, win 64240, options [mss 1460,sackOK,TS val 274177180
4 ecr 0,nop,wscale 7], length 0
09:40:32.412303 IP 10.0.2.15.42704 > 216.58.211.206.443: Flags [S], seq 4207187704, win 64240, options [mss 1460,sackOK,TS val 274177182
7 ecr 0,nop,wscale 7], length 0
09:40:32.447638 IP 216.58.211.206.443 > 10.0.2.15.42688: Flags [S.], seq 108416001, ack 3815115953, win 65535, options [mss 1460], lengt
h 0
09:40:32.447707 IP 10.0.2.15.42688 > 216.58.211.206.443: Flags [.], ack 1, win 64240, length 0
09:40:32.448724 IP 10.0.2.15.42688 > 216.58.211.206.443: Flags [P.], seq 1:1900, ack 1, win 64240, length 1899
09:40:32.449550 IP 216.58.211.206.443 > 10.0.2.15.42688: Flags [.], ack 1461, win 65535, length 0
09:40:32.449551 IP 216.58.211.206.443 > 10.0.2.15.42688: Flags [.], ack 1900, win 65535, length 0
09:40:32.469577 IP 216.58.211.206.443 > 10.0.2.15.42704: Flags [S.], seq 108480001, ack 4207187705, win 65535, options [mss 1460], lengt
```

Also can specify either if wanted as source port or distention port

```
$ sudo tcpdump -i enp0s3 -n dst port 211
```

- Logical operators : and - not - or

```
$ sudo tcpdump -i enp0s3 -n src 10.0.2.10 and dst port 3389
```

- Single quotes : are used to make it easier to combine big complex filters also using the brackets

```
sudo tcpdump -i enp0s3 'src 192.168.1.1 and dst 192.168.1.2 and not ( port 21 or 80 )'
```

- Protocols : we can filter using the protocol name ip tcp udp icmp arp ..... Etc

```
sudo tcpdump -i enp0s3 -n icmp
```

- Manual : see for more filters or usage

```
$ man tcpdump
```

- -w + path and name : to save or write the pcap file

```
$ sudo tcpdump -i enp0s3 -n host youtube.com -w Desktop/test.pcap
```

- -r + path and name : to read the pcap file

```
$ sudo tcpdump -r Desktop/test.pcap
```

- We can also make a filter on a saved pcap

```
m8a8a8y@m8a8a8y-VirtualBox:~$ sudo tcpdump -r Desktop/test.pcap port 443
reading from file Desktop/test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
10:05:46.588271 IP m8a8a8y-VirtualBox.53616 > dub08s02-in-f78.1e100.net.https: Flags [S], seq 3386377428, win 64240, options [mss 1460,sa
ackOK,TS val 3038340617 ecr 0,nop,wscale 7], length 0
10:05:46.594952 IP m8a8a8y-VirtualBox.53622 > dub08s02-in-f78.1e100.net.https: Flags [S], seq 877573133, win 64240, options [mss 1460,sa
ckOK,TS val 3038340623 ecr 0,nop,wscale 7], length 0
10:05:46.644521 IP dub08s02-in-f78.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [S.], seq 305984001, ack 3386377429, win 65535, opti
ons [mss 1460], length 0
```

## tcpdump: Analyzing Network Traffic

Tuesday, July 8, 2025 10:21 AM

- count : number of packets in a pcap

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sudo tcpdump -r test.pcap --count
reading from file test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
108 packets
```

- c num : will return the number of packets

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sudo tcpdump -r test.pcap -c 2
reading from file test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
10:05:46.580271 IP m8a8a8y-VirtualBox.53616 > dub88s02.in-f78.1e100.net.https: Flags [S], seq 3386377428, win 64240, options [mss 1460,sackOK,TS val 3038340617 ecr 0,nop,wscale 7], length 0
10:05:46.594952 IP m8a8a8y-VirtualBox.53622 > dub88s02.in-f78.1e100.net.https: Flags [S], seq 877573133, win 64240, options [mss 1460,sackOK,TS val 3038340623 ecr 0,nop,wscale 7], length 0
```

- t : will remove the time

But -tt and -tt -tttt all are format of time can be used

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sudo tcpdump -r test.pcap -t
reading from file test.pcap, link-type EN10MB (Ethernet), snapshot length 262144
IP m8a8a8y-VirtualBox.53616 > nrs09s08.in-f14.1e100.net.https: Flags [S], seq 3386377428
IP m8a8a8y-VirtualBox.53622 > nrs09s08.in-f14.1e100.net.https: Flags [S], seq 877573133
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [S.], seq 305904001
IP m8a8a8y-VirtualBox.53616 > nrs09s08.in-f14.1e100.net.https: Flags [.], ack 1, win 64240
IP m8a8a8y-VirtualBox.53616 > nrs09s08.in-f14.1e100.net.https: Flags [P.], seq 1:1897, ack 1:1897
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [.], ack 1461, win 64240
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53616: Flags [.], ack 1897, win 64240
IP nrs09s08.in-f14.1e100.net.https > m8a8a8y-VirtualBox.53622: Flags [S.], seq 306048001
IP m8a8a8y-VirtualBox.53622 > nrs09s08.in-f14.1e100.net.https: Flags [.], seq 1, win 64240
IP m8a8a8y-VirtualBox.53622 > nrs09s08.in-f14.1e100.net.https: Flags [P.], seq 1:1897, ack 1:1897
```

- Practical analysis : lets get some hands dirty

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop/02_Network_Security/01_tcpdump/PCAPs$ ls
2021-09-14.pcap 2024-04-18.pcap
```

Look for :

- the conversation number of packets
- See the most common IPs are talking
- See the port or the protocol being used
- Try to dig deeper example ( if its http try to look for get or post request or any file upload )

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop/02_Network_Security/01_tcpdump/PCAPs$ sudo tcpdump -r 2021-09-14.pcap --count
reading from file 2021-09-14.pcap, link-type EN10MB (Ethernet), snapshot length 262144
3679 packets
```

Most of the communication is between these 2 and its using http  
IP 10.0.0.168.49679 > 103.232.55.148.http:

We are trying to see if there is any file upload or download or website that we can see

```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop/02_Network_Security/01_tcpdump/PCAPs$ sudo tcpdump -r 2021-09-14.pcap host 10.0.0.168 and port 80
| grep -E "GET|POST"
```

```
6maxwidth=32768rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.035477 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 4294:4956, ack 5446, win 32768, length 662: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.201164 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 4956:5619, ack 6093, win 32608, length 663: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.407901 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 5619:6283, ack 6741, win 32606, length 664: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.703190 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 6283:6948, ack 7390, win 32768, length 665: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:31.860413 IP 10.0.0.168.49722 > 13.107.5.80.http: Flags [P.], seq 6948:7614, ack 8039, win 32606, length 666: HTTP: GET /qsnl.aspx?query=http3AN2FK2F103.232.55.148N2FserviceN2F.audlod
p&maxwidth=32768rowheight=20&sectionHeight=160&FORM=IES502&market=en-US HTTP/1.1
17:35:32.552578 IP 10.0.0.168.49724 > 103.232.55.148.http: Flags [P.], seq 1:282, ack 1, win 32768, length 281: HTTP: GET /servlce/.audlod.exe HTTP/1.1
17:36:44.610317 IP 10.0.0.168.49745 > geschken.on3cx.de.http: Flags [P.], seq 1:252, ack 1, win 256, length 251: HTTP: POST /-element/page.php?id=484 HTTP/1.0
17:36:45.103946 IP 10.0.0.168.49744 > geschken.on3cx.de.http: Flags [P.], seq 1:252, ack 1, win 256, length 251: HTTP: POST /-element/page.php?id=484 HTTP/1.0
17:36:45.679415 IP 10.0.0.168.49745 > geschken.on3cx.de.http: Flags [P.], seq 1:252, ack 1, win 256, length 251: HTTP: POST /-element/page.php?id=484 HTTP/1.0
17:36:55.102390 IP 10.0.0.168.49747 > 93.184.220.29.http: Flags [P.], seq 1:241, ack 1, win 256, length 248: HTTP: GET /MFEWtZBNMEsw5TA2BgUrDgMCqgA8BQ50otxN2Fh0ZtLk2Bz85IP17wEMvxdIQUTL3U1B
LV5uBu5q52F6K2BrK570VXj2KCEA177eL5ppmeL17jGdvdL0K3D HTTP/1.0
```

This file looks weird because it has (.) in the path this means a hidden file and if we look about the name of the file

It look a file form Microsoft



But its using http and is the IP its taken from for Microsoft ?  
Will no

**Quick Stats**

|              |                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------|
| IP Location  | Viet Nam Ho Noi Viettelserver Services Technology Company Limited                                     |
| ASN          | AS61737 VIETSERVER-AS-VN VIETSERVER SERVICES TECHNOLOGY COMPANY LIMITED, VN (registered Aug 25, 2014) |
| Whois Server | whois.apnic.net                                                                                       |
| IP Address   | 103.232.55.148                                                                                        |

🔍 Abuse contact for '103.232.52.0 - 103.232.55.255' is - [hs-charge@nmc.vn](mailto:hs-charge@nmc.vn)

```
inetnum: 103.232.52.0 - 103.232.55.255
netname: VIETSERVER-AS-VN
descr: VIETSERVER SERVICES TECHNOLOGY COMPANY LIMITED
descr: Xu Khuê, Chu Phan ward, Me Linh district, Ha Noi city
country: VN
tech-c: NWS2-AP
org-c: VN
mnt-by: MAINT-VN-VNNIC
aut-c: IRT-VNNIC-AP
status: ALLOCATED PORTABLE
mnt-lf: NWS2-AP
mnt-rf: 2013-09-26T09:36:00Z
```

```
m0le@kali:~/m0le$ sudo tcpdump -r 2021-09-14.pcap | grep "audiolog.exe"
```

Less is to make it easy to scroll

```

17:35:31.040413 IP 10.0.0.168.49722 → [13.107.5.80:http: [Flags [P]], seq 6940:7614, ack 8039, win 32686, length 666: HTTP: GET /qsn1.aspx?query=http%3A%2F%2F103.232.25.148%2Fservice%2F.exe&maxwidth=32765&rowheight=20&sectionheight=160&FORM=IESS02&market=en-US HTTP/1.1
...
...k.P:1:PS:1.t:JP....y..GET /qsn1.aspx?query=http%3A%2F%2F103.232.25.148%2Fservice%2F.exe&maxwidth=32765&rowheight=20&sectionheight=160&FORM=IESS02&market=en-US HTTP/1.1
Accept: */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64; Trident/7.0; rv:11.0) like Gecko
Accept-Encoding: gzip, deflate
Host: api.bing.com
Connection: Keep-Alive
Cookie: MUIDB=3B126A8F5BAE6E739BA7A3D590D6F45; _EDGE_S=F&SID=38BF9AD3FD064171AE8BA61FC1E656E; _EDGE_V=1; SRCHD=AF=IESS02; SRCHUIO=V=Q&ZUID=2B8531E85C146A6B0512C61CE102958&nncg=1; SRCHSRDOB=20210914; SRCHPGUSR=SRCHLANG=en; _SS=SID=38BF9AD3FD064171AE8BA61FC1E656E; MUIDB=3B126A8F5BAE6E739BA7A3D590D6F45

17:35:31.042195 IP 13.107.5.80.http → 10.0.0.168.49722: Flags [..], ack 7614, win 16419, length 0
E(...@.t..H.k.P
...P:1:t:1:J5.MUP.0@0...
17:35:31.098207 IP 13.107.5.80.http → 10.0.0.168.49722: Flags [P..], seq 8039:8688, ack 7614, win 16419, length 649: HTTP: HTTP/1.1 200 OK
E(...@.t..k..P
....P:1:t:1:J5.MUP.0@...HTTP/1.1 200 OK
Cache-Control: no-cache, no-store, must-revalidate
Pragma: no-cache
Content-Length: 192
Content-Type: text/html; charset=utf-8
Content-Encoding: gzip
Expires: -1
Vary: Accept-Encoding
P3P: CP="NON UNI COM NAV STA LOC CURA DEVA PSAA PSDa OUR IND"
K-SMR-Routing: 1
K-Cache: CONFIG_NOCACHE
K-MSEdge-Ref: Ref: 04B49C8ABA044DF78E2B469C8B55C5C9 Ref B: AM504EDGE1622 Ref C: 2021-09-14T14:35:33Z
Date: Tue, 14 Sep 2021 14:35:31 GMT
*****

```

This url looks suspicious ( don't forget to add defang )

The screenshot shows the Burp Suite interface. On the left, the 'Recipe' tab is active, displaying a 'URL Decode' button and a checked checkbox labeled 'Treat "\*" as space'. On the right, the 'Input' field contains the URL 'http://103.232.55.148/service/2F.audiog.exe'.

The screenshot shows the VirusShare website interface. At the top, the URL `http://103.232.55.148/service.audodg.me` is entered in the search bar. The main content area features a circular security score of **4/57** with a red 'Malware' status indicator. Below this, a table titled 'Security vendors' analysis' provides detailed detection results:

| Security vendor | Analysis | Result  |
|-----------------|----------|---------|
| BitDefender     | Malware  | Malware |
| G-Data          | Webroot  | Malware |
| Avast           | Clean    | Clean   |

The interface also includes a 'Join our Community' banner and a 'Last Analysis Date' of 23 days ago.

IP Location  United States Redmond Microsoft Corporation  
 ASN  AS8068 Microsoft Corporation ORP-MSN-AS-BLOCK, US [registered Mar 31, 1997]  
 Whois Server whois.a2n.net  
 IP Address 13.107.5.80  
 Reverse IP 1 website uses this address.







# tcpdump Challenge 1

Thursday, July 10, 2025

1:01 PM

## Challenge URL

- <https://challenges.malwarecube.com/#/c/bb2d4b38-3f4f-4639-8b09-ef17bb229b33>

# Introduction to Wireshark

Wednesday, July 9, 2025 9:23 AM

# Wireshark: Capture and Display Filters

Wednesday, July 9, 2025 10:26 AM

Capture filter is used to make less packet and traffic capture -----> for more info look at <https://wiki.wireshark.org/CaptureFilters>

Welcome to Wireshark

## Capture

...using this filter: ☒ host google.com and (port 80)

- enp0s3
- any
- Loopback: lo
- bluetooth-monitor
- nflag
- nqueue
- dbus-system
- dbus-session
- Cisco remote capture: ciscodump
- DisplayPort AUX channel monitor capture: dpauxmon
- Random packet generator: randpkt
- systemd Journal Export: sdjournal
- SSH remote capture: sshdump
- UDP Listener remote capture: udpdump
- Wi-Fi remote capture: wifidump

Display filter is filtering the packet been already captured -----> for more info look at <https://wiki.wireshark.org/DisplayFilters>

| No. | Time         | Source    | Destination | Protocol | Length | Info                                                                                                           |
|-----|--------------|-----------|-------------|----------|--------|----------------------------------------------------------------------------------------------------------------|
| 643 | 11.998661934 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519269544 TSecr=0 WS=128                      |
| 644 | 12.004333547 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519269550 TSecr=0 WS=128                      |
| 645 | 12.257475204 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519269803 TSecr=0 WS=128                      |
| 646 | 13.014456691 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519270560 TSecr=0 WS=128 |
| 647 | 13.014624191 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519270560 TSecr=0 WS=128 |
| 648 | 13.270263023 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519270816 TSecr=0 WS=128 |
| 649 | 14.037893638 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519271583 TSecr=0 WS=128 |
| 650 | 14.037251826 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519271583 TSecr=0 WS=128 |
| 651 | 14.294180179 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519271840 TSecr=0 WS=128 |
| 652 | 15.002093658 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519272608 TSecr=0 WS=128 |
| 653 | 15.002250236 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519272608 TSecr=0 WS=128 |
| 654 | 15.317579257 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519272803 TSecr=0 WS=128 |
| 655 | 16.006737180 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519273632 TSecr=0 WS=128 |
| 656 | 16.007112947 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519273633 TSecr=0 WS=128 |
| 657 | 16.341421701 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519273887 TSecr=0 WS=128 |
| 662 | 17.100909541 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44174 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519274655 TSecr=0 WS=128 |
| 663 | 17.109085529 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519274655 TSecr=0 WS=128 |
| 664 | 17.365202177 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44180 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519274911 TSecr=0 WS=128 |
| 665 | 19.150114400 | 10.0.2.15 | 8.8.8.8     | TCP      | 74     | [TCP Retransmission] 44164 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1519276704 TSecr=0 WS=128 |

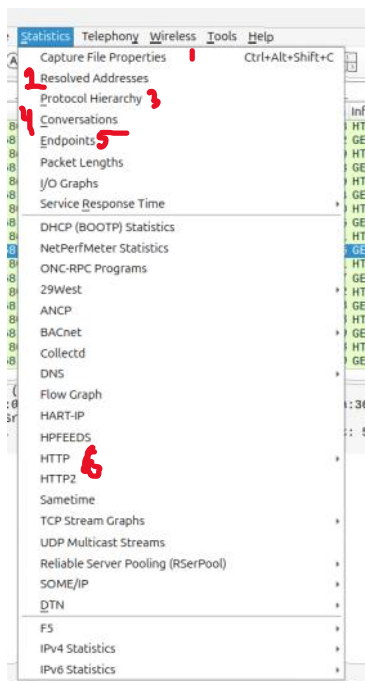
In display filter we can show different options

- IPs as wanted
- Ports
- Text or some words

| No.  | Time       | Source         | Destination    | Protocol | Length | Info                                                                                                                                  |
|------|------------|----------------|----------------|----------|--------|---------------------------------------------------------------------------------------------------------------------------------------|
| 2642 | 194.757055 | 10.0.0.168     | 13.107.5.80    | HTTP     | 710    | GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IESS02&mark |
| 2644 | 194.811822 | 13.107.5.80    | 10.0.0.168     | HTTP/X.. | 701    | HTTP/1.1 200 OK                                                                                                                       |
| 2646 | 195.002742 | 10.0.0.168     | 13.107.5.80    | HTTP     | 717    | GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IESS02&mar  |
| 2648 | 195.060797 | 13.107.5.80    | 10.0.0.168     | HTTP/X.. | 702    | HTTP/1.1 200 OK                                                                                                                       |
| 2651 | 195.219559 | 10.0.0.168     | 13.107.5.80    | HTTP     | 718    | GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.e&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IESS02&ma  |
| 2653 | 195.287168 | 13.107.5.80    | 10.0.0.168     | HTTP/X.. | 703    | HTTP/1.1 200 OK                                                                                                                       |
| 2655 | 195.424768 | 10.0.0.168     | 13.107.5.80    | HTTP     | 719    | GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.ex&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IESS02&m  |
| 2657 | 195.480048 | 13.107.5.80    | 10.0.0.168     | HTTP/X.. | 703    | HTTP/1.1 200 OK                                                                                                                       |
| 2659 | 195.561991 | 10.0.0.168     | 13.107.5.80    | HTTP     | 720    | GET /qsm.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiiodg.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IESS02&  |
| 2661 | 195.619865 | 13.107.5.80    | 10.0.0.168     | HTTP/X.. | 703    | HTTP/1.1 200 OK                                                                                                                       |
| 2677 | 196.274156 | 10.0.0.168     | 103.232.55.148 | HTTP     | 335    | GET /service/.audiiodg.exe HTTP/1.1                                                                                                   |
| 3421 | 198.208642 | 103.232.55.148 | 10.0.0.168     | HTTP     | 900    | HTTP/1.1 200 OK (application/x-msdownload)                                                                                            |

# Wireshark: Statistics

Wednesday, July 9, 2025 10:52 AM



Wireshark - Capture File Properties - 2021-09-14.pcap

Details

**File**

Name: /home/m8a8a8y/Desktop/02\_Network\_Security/02\_Wireshark/PCAPs/2021-09-14.pcap  
Length: 2,450 kB  
Hash (SHA256): 6d0c4c0b58274b25d0711f7ce29ddc8d7c0c6e594d73bddc2a1e2a90923d1c87  
Hash (SHA1): c2fa875fe4618d9931fe8b05e1c10ec8e3b5a6b6  
Format: Wireshark/cpdump/... - pcap  
Encapsulation: Ethernet  
Snapshot length: 262144

**Time**

First packet: 2021-09-14 17:32:16  
Last packet: 2021-09-14 17:42:01  
Elapsed: 00:09:44

**Capture**

Hardware: Unknown  
OS: Unknown  
Application: Unknown

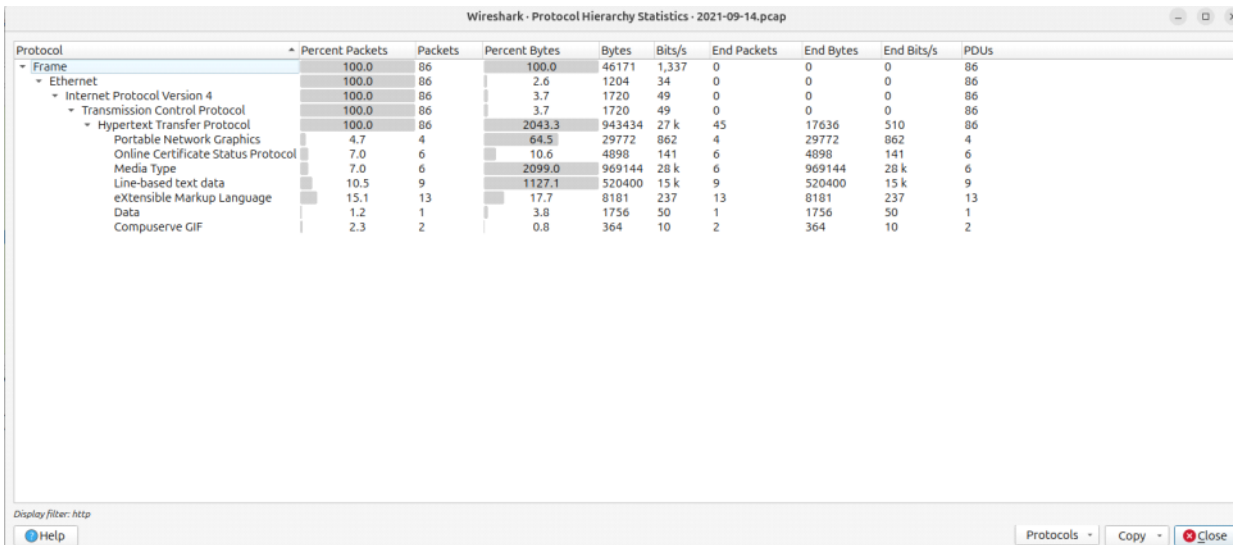
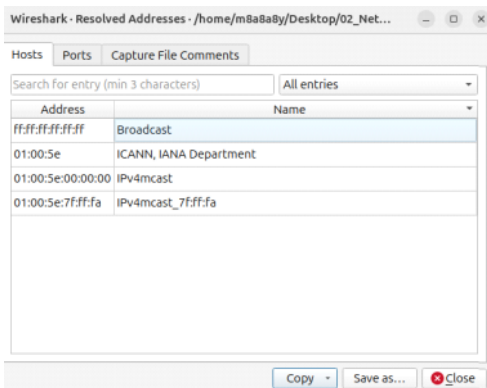
**Interfaces**

| Interface | Dropped packets | Capture filter | Link type | Packet size limit (snaplen) |
|-----------|-----------------|----------------|-----------|-----------------------------|
| Unknown   | Unknown         | Unknown        | Ethernet  | 262144 bytes                |

**Statistics**

| Measurement            | Captured | Displayed    | Marked |
|------------------------|----------|--------------|--------|
| Packets                | 3679     | 86 (2.3%)    | —      |
| Time span, s           | 584.905  | 276.117      | —      |
| Average pps            | 6.3      | 0.3          | —      |
| Average packet size, B | 650      | 537          | —      |
| Bytes                  | 2391470  | 46171 (1.9%) | 0      |
| Average bytes/s        | 4,088    | 167          | —      |
| Average bits/s         | 32 k     | 1,337        | —      |

Help Refresh Edit Comments Copy To Clipboard Close



Wireshark · Conversations - 2021-09-14.pcap

| Ethernet - 1 | IPv4 - 10      | IPv6    | TCP - 24  | UDP       |               |                  |               |             |               |             |            |          |              |          |
|--------------|----------------|---------|-----------|-----------|---------------|------------------|---------------|-------------|---------------|-------------|------------|----------|--------------|----------|
| Address A ▾  | Address B      | Packets | Bytes     | Stream ID | Total Packets | Percent Filtered | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start  | Duration | Bits/s A → B | Bits/s B |
| 10.0.0.168   | 13.107.5.80    | 24      | 16 kB     | 18        | 60            | 40.00%           | 12            | 8 kB        | 12            | 8 kB        | 191.783934 | 71.6920  | 921 bits/s   | 878 b    |
| 10.0.0.168   | 23.209.125.71  | 2       | 1 kB      | 6         | 14            | 14.29%           | 1             | 299 bytes   | 1             | 843 bytes   | 31.497565  | 109.9813 | 21 bits/s    | 61 b     |
| 10.0.0.168   | 31.13.64.21    | 2       | 423 bytes | 16        | 158           | 1.27%            | 1             | 361 bytes   | 1             | 62 bytes    | 157.902150 | 64.2260  | 44 bits/s    | 7 b      |
| 10.0.0.168   | 69.16.175.10   | 2       | 438 bytes | 15        | 70            | 2.86%            | 1             | 364 bytes   | 1             | 74 bytes    | 154.773375 | 67.3575  | 43 bits/s    | 8 b      |
| 10.0.0.168   | 72.21.81.240   | 2       | 681 bytes | 13        | 10            | 20.00%           | 1             | 336 bytes   | 1             | 345 bytes   | 92.170463  | 60.2113  | 44 bits/s    | 45 b     |
| 10.0.0.168   | 93.184.220.29  | 12      | 6 kB      | 7         | 52            | 23.08%           | 6             | 2 kB        | 6             | 4 kB        | 31.550561  | 381.1171 | 36 bits/s    | 89 b     |
| 10.0.0.168   | 103.232.55.148 | 32      | 18 kB     | 2         | 2,075         | 1.54%            | 16            | 6 kB        | 16            | 12 kB       | 2.434030   | 219.9713 | 210 bits/s   | 447 b    |
| 10.0.0.168   | 104.16.18.94   | 2       | 1 kB      | 14        | 23            | 8.70%            | 1             | 382 bytes   | 1             | 816 bytes   | 154.521763 | 67.6126  | 45 bits/s    | 96 b     |
| 10.0.0.168   | 104.85.5.163   | 2       | 1 kB      | 12        | 11            | 18.18%           | 1             | 215 bytes   | 1             | 812 bytes   | 68.008696  | 73.4689  | 23 bits/s    | 88 b     |
| 10.0.0.168   | 136.243.159.53 | 6       | 1 kB      | 19        | 36            | 16.67%           | 3             | 853 bytes   | 3             | 162 bytes   | 268.315691 | 1.4308   | 4,769 bits/s | 905 b    |

Close



Wireshark - Endpoints - 2021-09-14.pcap

| Ethernet - 2   |         | IPv4 - 11 |               | IPv6             |            | TCP - 34  |            | UDP       |         |      |          |           |           |                 |  |  |  |  |  |
|----------------|---------|-----------|---------------|------------------|------------|-----------|------------|-----------|---------|------|----------|-----------|-----------|-----------------|--|--|--|--|--|
| Address        | Packets | Bytes     | Total Packets | Percent Filtered | Tx Packets | Tx Bytes  | Rx Packets | Rx Bytes  | Country | City | Latitude | Longitude | AS Number | AS Organization |  |  |  |  |  |
| 10.0.0.168     | 86      | 46 kB     | 3,649         | 2.36%            | 43         | 19 kB     | 43         | 28 kB     |         |      |          |           |           |                 |  |  |  |  |  |
| 13.107.5.80    | 24      | 16 kB     | 60            | 40.00%           | 12         | 8 kB      | 12         | 8 kB      |         |      |          |           |           |                 |  |  |  |  |  |
| 23.209.125.71  | 2       | 1 kB      | 14            | 14.29%           | 1          | 843 bytes | 1          | 299 bytes |         |      |          |           |           |                 |  |  |  |  |  |
| 31.13.64.21    | 2       | 423 bytes | 158           | 1.27%            | 1          | 62 bytes  | 1          | 361 bytes |         |      |          |           |           |                 |  |  |  |  |  |
| 69.16.175.10   | 2       | 438 bytes | 70            | 2.86%            | 1          | 74 bytes  | 1          | 364 bytes |         |      |          |           |           |                 |  |  |  |  |  |
| 72.21.81.240   | 2       | 681 bytes | 10            | 20.00%           | 1          | 345 bytes | 1          | 336 bytes |         |      |          |           |           |                 |  |  |  |  |  |
| 93.184.220.29  | 12      | 6 kB      | 52            | 23.08%           | 6          | 4 kB      | 6          | 2 kB      |         |      |          |           |           |                 |  |  |  |  |  |
| 103.232.55.148 | 32      | 18 kB     | 2,075         | 1.54%            | 16         | 12 kB     | 16         | 6 kB      |         |      |          |           |           |                 |  |  |  |  |  |
| 104.16.18.94   | 2       | 1 kB      | 23            | 8.70%            | 1          | 816 bytes | 1          | 382 bytes |         |      |          |           |           |                 |  |  |  |  |  |
| 104.85.5.163   | 2       | 1 kB      | 11            | 18.18%           | 1          | 812 bytes | 1          | 215 bytes |         |      |          |           |           |                 |  |  |  |  |  |
| 136.243.159.53 | 6       | 1 kB      | 36            | 16.67%           | 3          | 162 bytes | 3          | 853 bytes |         |      |          |           |           |                 |  |  |  |  |  |

Close

HPFEEDS :: 5446, Len: 602

- HTTP
  - Packet Counter
  - Requests
  - Load Distribution
  - Request Sequences
- HTTP2
- Sametime
- TCP Stream Graphs
- UDP Multicast Streams

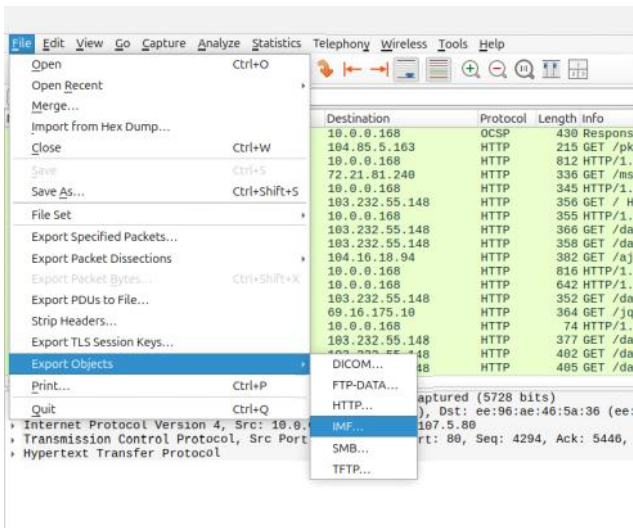
Wireshark - HTTP / Requests - 2021-09-14.pcap

| Request Type                                                                                                                                | Count | Average | Min Val | Max Val | Rate (ms) | Percent | Burst Rate | Burst Start |
|---------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|---------|---------|-----------|---------|------------|-------------|
| /MEFWT2BNMEswSTAJBgUrDgMCCGUABBBQ500tx%2Fh0Zt%2Bz8SiPi7wEwVxDIQUtUUIBIV5uNuSg%2F6%2BkS7QYXjzCEA177eIggmWeUjG4vdGL0%3D                       | 1     | 0.0000  | 16.67%  | 0.0100  | 278.824   |         |            |             |
| ctdl.windowsupdate.com                                                                                                                      | 2     | 0.0000  | 3.92%   | 0.0100  | 31.500    |         |            |             |
| /msdownload/update/v3/static/trusted/en/pinrulesLcab7f0c731147f2df6                                                                         | 1     | 0.0000  | 50.00%  | 0.0100  | 92.242    |         |            |             |
| /msdownload/update/v3/static/trusted/en/disallowedcertstLcab73027e92ff72bd024                                                               | 1     | 0.0000  | 50.00%  | 0.0100  | 31.500    |         |            |             |
| connect.facebook.net                                                                                                                        | 1     | 0.0000  | 1.96%   | 0.0100  | 157.904   |         |            |             |
| /en_US/all.js                                                                                                                               | 1     | 0.0000  | 100.00% | 0.0100  | 157.904   |         |            |             |
| code.jquery.com                                                                                                                             | 1     | 0.0000  | 1.96%   | 0.0100  | 154.776   |         |            |             |
| /query-1.10.2.min.js                                                                                                                        | 1     | 0.0000  | 100.00% | 0.0100  | 154.776   |         |            |             |
| cdnjs.cloudflare.com                                                                                                                        | 1     | 0.0000  | 1.96%   | 0.0100  | 154.527   |         |            |             |
| /ajax/libs/font-awesome/3.1.0/css/font-awesome.min.css                                                                                      | 1     | 0.0000  | 100.00% | 0.0100  | 154.527   |         |            |             |
| api.bing.com                                                                                                                                | 12    | 0.0000  | 23.53%  | 0.0100  | 191.787   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 195.562   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 195.425   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 195.220   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 195.003   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 194.757   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 194.589   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 193.023   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 192.811   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 192.661   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 192.259   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 192.006   |         |            |             |
| /qsl.aspx?query=http%3A%2F%2F103.232.55.148%2Fservice%2F.audiogd.exe&maxwidth=32765&rowheight=20&sectionHeight=160&FORM=IES502&market=en-US | 1     | 0.0000  | 8.33%   | 0.0100  | 191.787   |         |            |             |
| 239.255.255.250:1900                                                                                                                        | 8     | 0.0000  | 15.69%  | 0.0200  | 57.883    |         |            |             |
| *                                                                                                                                           | 8     | 0.0000  | 100.00% | 0.0200  | 57.883    |         |            |             |
| 136.243.159.53                                                                                                                              | 3     | 0.0000  | 5.88%   | 0.0100  | 268.358   |         |            |             |
| /element/page.php?id=484                                                                                                                    | 3     | 0.0000  | 100.00% | 0.0100  | 268.358   |         |            |             |
| 103.232.55.148                                                                                                                              | 16    | 0.0001  | 31.37%  | 0.0400  | 154.752   |         |            |             |
| /service/audiogd.exe                                                                                                                        | 1     | 0.0000  | 6.25%   | 0.0100  | 196.274   |         |            |             |
| /service/                                                                                                                                   | 1     | 0.0000  | 6.25%   | 0.0100  | 2.708     |         |            |             |
| /icons/blank.gif                                                                                                                            | 1     | 0.0000  | 6.25%   | 0.0100  | 3.164     |         |            |             |
| /icons/back.gif                                                                                                                             | 1     | 0.0000  | 6.25%   | 0.0100  | 3.165     |         |            |             |
| /favicon.ico                                                                                                                                | 1     | 0.0000  | 6.25%   | 0.0100  | 3.864     |         |            |             |
| /dashboard/stylesheets/normalize.css                                                                                                        | 1     | 0.0000  | 6.25%   | 0.0100  | 154.496   |         |            |             |
| /dashboard/stylesheets/all.css                                                                                                              | 1     | 0.0000  | 6.25%   | 0.0100  | 154.752   |         |            |             |
| /dashboard/javascripts/modernizr.js                                                                                                         | 1     | 0.0000  | 6.25%   | 0.0100  | 154.785   |         |            |             |
| /dashboard/javascripts/all.js                                                                                                               | 1     | 0.0000  | 6.25%   | 0.0100  | 155.031   |         |            |             |
| /dashboard/images/kampp-logo.svg                                                                                                            | 1     | 0.0000  | 6.25%   | 0.0100  | 154.791   |         |            |             |
| /dashboard/images/social-icons.png                                                                                                          | 1     | 0.0000  | 6.25%   | 0.0100  | 158.046   |         |            |             |

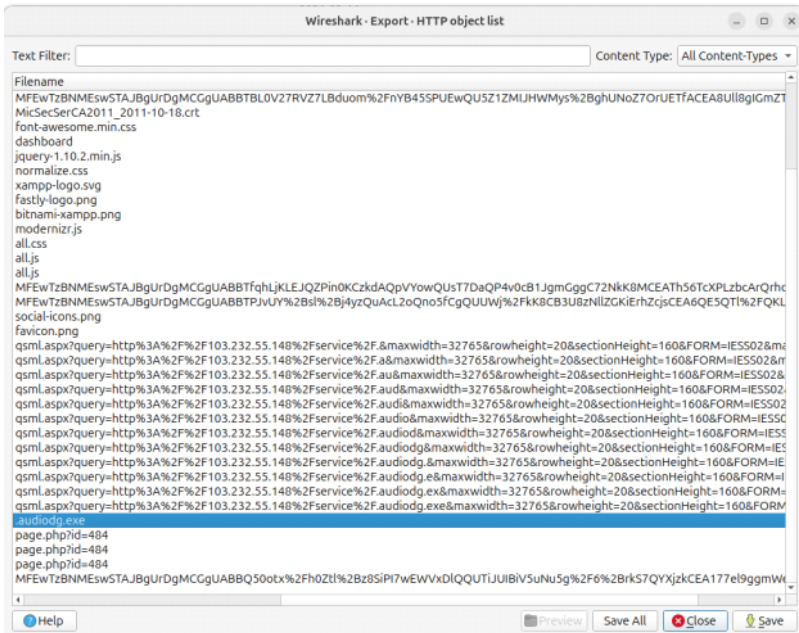
Remember it? ☺

Display filter: Enter a display filter ...

Copy Save as... Close



Form here u can install anything ( take the hash or file type and see on viruse total



```
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ ls -la
.. 01_Phishing_Analysis 02_Network_Security 03_Endpoint_Security 04_SIEM 05_Threat_Intelligence 06_Digital_Forensics 07_Incident_Response .audioiodg.exe SOC
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ file .audioiodg.exe
.audioiodg.exe: PE32 executable (GUI) Intel 80386 Mono/.Net assembly, for MS Windows, 3 sections
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$ sha256sum .audioiodg.exe
f485d1a65ccf9f857baa49725d337c15e8aa34515b85c8ef59a72afad7b85249 .audioiodg.exe
m8a8a8y@m8a8a8y-VirtualBox: ~/Desktop$
```

58

72

Community Score

58/72 security vendors flagged this file as malicious

ReanalyzeSimilarMore

f485d1a65ccf9f857baa49725d337c15e8aa34515b85c8ef59a72afad7b85249

Size586.50 KBLast Analysis Date1 month ago

CallerFilePathAttribu.exe

peaseassemblyruntime-modulesdirect-cpu-clock-accesschecks-user-inputspreaderdetect-debug-environmentpersistence-long-sleepsmalware

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY1

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat labeltrojan.msi/agenttesla

Threat categoriestrojan

Family labelsmsiagentteslaagensia

Security vendors' analysis

Do you want to automate checks?

|                  |                                        |             |                                    |
|------------------|----------------------------------------|-------------|------------------------------------|
| AhnLab-V3        | Malware/Win.Generic.C4631799           | Alibaba     | TrojanPSW:MSIL/AgentTesla.c542bc5b |
| AliCloud         | Trojan(stealer):MSIL/AgentTesla.SZ8PHU | ALYac       | Spyware.LokiBot                    |
| Arcabit          | Trojan.Generic.D2CCB080                | Arctic Wolf | Unsafe                             |
| Avast            | Win32:MalwareX-gen [Rat]               | AVG         | Win32:MalwareX-gen [Rat]           |
| Avira (no cloud) | HEUR/AGEN.1324055                      | BitDefender | Trojan.GenericKD.46968960          |

Network Security Page 21

# Wireshark: Analyzing Network Traffic

Thursday, July 10, 2025 12:18 PM

Look for :

- the conversation number of packets
- See the most common IPs are talking
- See the port or the protocol being used
- Try to dig deeper example ( if its http try to look for get or post request or any file upload )

File

Name: /home/m8a8a8y/Desktop/02\_Network\_Security/02\_Wireshark/PCAPs/2023-02-03.pcap  
Length: 32 MB  
Hash (SHA256): c071c04241b96b91da5dad98fc1fc57da47227e2bbd8165a314c980f55a948d  
Hash (SHA1): f9510a9b1e336eafcd643af65fc3a4ba059b478a  
Format: Wireshark/tcpdump/... - pcap  
Encapsulation: Ethernet  
Snapshot length: 65535

Time

First packet: 2023-02-03 20:02:36  
Last packet: 2023-02-03 22:54:43  
Elapsed: 02:52:06

Capture

Hardware: Unknown  
OS: Unknown  
Application: Unknown

Interfaces

Interface

Unknown

Dropped packets

Unknown

Capture filter

Unknown

Link type

Ethernet

Packet size limit (snaplen)

65535 bytes

Statistics

Measurement

Captured

Displayed

Marked

Packets 55207 (100.0%)  
Time span, s 10326.359  
Average pps 5.3  
Average packet size, B 565  
Bytes 31172531  
Average bytes/s 3,018  
Average bits/s 24 k

| Ethernet II | IP v4 - 150     | IPv6    | TCP - 762 | UDP - 722 |               |             |               |             |             |            |              |              |  |  |
|-------------|-----------------|---------|-----------|-----------|---------------|-------------|---------------|-------------|-------------|------------|--------------|--------------|--|--|
| Address A   | Address B       | Packets | Bytes     | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start   | Duration   | Bits/s A → B | Bits/s B → A |  |  |
| 10.0.0.149  | 10.0.0.6        | 12,040  | 8 MB      | 5         | 8,036         | 7 MB        | 4,004         | 1 MB        | 0.184672    | 10295.0052 | 5,115 bits/s | 829 bits/s   |  |  |
| 10.0.0.149  | 208.187.122.14  | 2,860   | 7 MB      | 39        | 4,712         | 474 kB      | 5,094         | 6 MB        | 695.938749  | 9430.4205  | 402 bits/s   | 5,392 bits/s |  |  |
| 10.0.0.149  | 78.31.67.7      | 3,913   | 439 kB    | 49        | 1,947         | 311 kB      | 1,966         | 128 kB      | 1335.473325 | 8972.0715  | 277 bits/s   | 114 bits/s   |  |  |
| 10.0.0.149  | 5.75.205.43     | 4,127   | 3 MB      | 96        | 1,878         | 3 MB        | 2,249         | 124 kB      | 8328.021113 | 21.8323    | 1,005 kbps   | 45 kbps      |  |  |
| 10.0.0.149  | 74.6.143.26     | 2,747   | 2 MB      | 86        | 1,314         | 75 kB       | 1,433         | 2 MB        | 5780.995379 | 3027.0978  | 197 bits/s   | 5,244 bits/s |  |  |
| 10.0.0.149  | 23.111.114.52   | 2,342   | 335 kB    | 98        | 1,071         | 235 kB      | 1,271         | 100 kB      | 8396.331884 | 1242.2290  | 1,514 bits/s | 642 bits/s   |  |  |
| 10.0.0.149  | 13.107.42.14    | 1,449   | 1 MB      | 63        | 674           | 53 kB       | 775           | 979 kB      | 2537.471296 | 6677.8955  | 63 bits/s    | 1,173 bits/s |  |  |
| 10.0.0.149  | 128.254.207.55  | 1,924   | 2 MB      | 25        | 639           | 35 kB       | 1,285         | 2 MB        | 106.401425  | 2.2173     | 124 kbps     | 6,605 kbps   |  |  |
| 10.0.0.149  | 23.58.117.164   | 1,020   | 722 kB    | 76        | 497           | 32 kB       | 523           | 690 kB      | 4253.589506 | 5197.2095  | 49 bits/s    | 1,062 bits/s |  |  |
| 10.0.0.149  | 102.156.32.143  | 892     | 680 kB    | 31        | 471           | 655 kB      | 421           | 25 kB       | 446.974674  | 433.6001   | 12 kbps      | 458 bits/s   |  |  |
| 10.0.0.149  | 209.131.162.45  | 1,047   | 710 kB    | 55        | 470           | 34 kB       | 577           | 677 kB      | 1583.379429 | 7224.7145  | 37 bits/s    | 749 bits/s   |  |  |
| 10.0.0.149  | 23.214.54.85    | 877     | 657 kB    | 99        | 411           | 28 kB       | 466           | 629 kB      | 8515.268334 | 1316.3443  | 168 bits/s   | 3,823 bits/s |  |  |
| 10.0.0.149  | 69.58.187.40    | 880     | 650 kB    | 78        | 400           | 25 kB       | 480           | 626 kB      | 4446.029815 | 4362.0639  | 45 bits/s    | 1,147 bits/s |  |  |
| 10.0.0.149  | 184.86.169.24   | 764     | 550 kB    | 59        | 375           | 26 kB       | 389           | 524 kB      | 1964.661047 | 329.5618   | 624 bits/s   | 12 kbps      |  |  |
| 10.0.0.149  | 20.10.31.115    | 1,024   | 596 kB    | 10        | 350           | 40 kB       | 674           | 557 kB      | 55.493103   | 10216.1588 | 31 bits/s    | 436 bits/s   |  |  |
| 10.0.0.149  | 239.255.255.250 | 317     | 68 kB     | 21        | 317           | 68 kB       | 0             | 0 bytes     | 57.853976   | 10215.0031 | 53 bits/s    | 0 bits/s     |  |  |
| 10.0.0.149  | 23.64.146.226   | 524     | 376 kB    | 65        | 244           | 19 kB       | 280           | 357 kB      | 2730.381668 | 4715.2357  | 32 bits/s    | 605 bits/s   |  |  |
| 10.0.0.149  | 96.6.184.69     | 404     | 282 kB    | 95        | 194           | 14 kB       | 210           | 267 kB      | 8071.029433 | 1599.8243  | 70 bits/s    | 1,337 bits/s |  |  |
| 10.0.0.149  | 142.250.68.132  | 295     | 151 kB    | 48        | 145           | 16 kB       | 150           | 134 kB      | 1331.027958 | 8881.4026  | 14 bits/s    | 121 bits/s   |  |  |
| 10.0.0.149  | 184.51.39.62    | 269     | 187 kB    | 30        | 129           | 9 kB        | 140           | 179 kB      | 445.527905  | 655.1113   | 107 bits/s   | 2,179 bits/s |  |  |
| 10.0.0.149  | 72.163.4.185    | 238     | 78 kB     | 29        | 117           | 17 kB       | 121           | 61 kB       | 444.801479  | 9084.5539  | 15 bits/s    | 53 bits/s    |  |  |
| 10.0.0.149  | 96.7.169.183    | 248     | 180 kB    | 45        | 116           | 7 kB        | 132           | 173 kB      | 1138.770324 | 109.6244   | 535 bits/s   | 12 kbps      |  |  |
| 10.0.0.149  | 23.210.67.195   | 234     | 166 kB    | 80        | 114           | 7 kB        | 120           | 159 kB      | 4638.368628 | 109.6052   | 498 bits/s   | 11 kbps      |  |  |
| 10.0.0.149  | 104.93.55.25    | 225     | 143 kB    | 92        | 105           | 16 kB       | 120           | 127 kB      | 7115.973098 | 1472.1700  | 87 bits/s    | 689 bits/s   |  |  |
| 10.0.0.149  | 138.1.33.162    | 157     | 52 kB     | 50        | 81            | 12 kB       | 76            | 39 kB       | 1394.941651 | 8900.1963  | 10 bits/s    | 35 bits/s    |  |  |
| 10.0.0.149  | 72.247.204.96   | 155     | 93 kB     | 52        | 77            | 8 kB        | 78            | 85 kB       | 1395.397019 | 1059.9605  | 58 bits/s    | 643 bits/s   |  |  |
| 10.0.0.149  | 40.126.28.18    | 139     | 48 kB     | 28        | 76            | 11 kB       | 63            | 37 kB       | 372.306497  | 9720.6517  | 8 bits/s     | 30 bits/s    |  |  |

See what protocols are mostly used

| Protocol                                                            | Percent Packets | Packets | Percent Bytes | Bytes    | Bits/s | End Packets | End Bytes | End Bits/s | POUs  |
|---------------------------------------------------------------------|-----------------|---------|---------------|----------|--------|-------------|-----------|------------|-------|
| Frame                                                               | 100.0           | 55207   | 100.0         | 31172331 | 24 k   | 0           | 0         | 0          | 55207 |
| Ethernet                                                            | 100.0           | 55207   | 2.5           | 772898   | 598    | 0           | 0         | 0          | 55207 |
| Internet Protocol Version 4                                         | 96.3            | 53164   | 3.4           | 1063308  | 823    | 0           | 0         | 0          | 53164 |
| User Datagram Protocol                                              | 3.1             | 1730    | 0.0           | 13840    | 10     | 0           | 0         | 0          | 1730  |
| Simple Service Discovery Protocol                                   | 0.6             | 317     | 0.2           | 54974    | 42     | 317         | 54974     | 42         | 317   |
| Remote Procedure Call                                               | 0.0             | 3       | 0.0           | 228      | 0      | 0           | 0         | 0          | 3     |
| Portmap                                                             | 0.0             | 3       | 0.0           | 48       | 0      | 3           | 48        | 0          | 3     |
| Network Time Protocol                                               | 0.0             | 22      | 0.0           | 2640     | 2      | 22          | 2640      | 2          | 22    |
| NetBIOS Name Service                                                | 0.1             | 29      | 0.0           | 1810     | 1      | 29          | 1810      | 1          | 29    |
| NetBIOS Datagram Service                                            | 0.1             | 48      | 0.0           | 3936     | 3      | 0           | 0         | 0          | 48    |
| Multicast Domain Name System                                        | 0.0             | 11      | 0.0           | 448      | 0      | 11          | 448       | 0          | 11    |
| Link-local Multicast Name Resolution                                | 0.0             | 1       | 0.0           | 33       | 0      | 1           | 33        | 0          | 1     |
| Dynamic Host Configuration Protocol                                 | 0.0             | 6       | 0.0           | 1879     | 1      | 6           | 1879      | 1          | 6     |
| Domain Name System                                                  | 1.9             | 1061    | 0.3           | 85490    | 66     | 1061        | 85490     | 66         | 1061  |
| Connectionless Lightweight Directory Access Protocol                | 0.4             | 232     | 0.1           | 45980    | 35     | 232         | 45980     | 35         | 232   |
| Transmission Control Protocol                                       | 93.1            | 51423   | 3.4           | 1046060  | 810    | 38837       | 794340    | 615        | 51423 |
| Transport Layer Security                                            | 13.2            | 7277    | 62.2          | 19375514 | 15 k   | 7277        | 17220336  | 13 k       | 7724  |
| Simple Mail Transfer Protocol                                       | 0.1             | 37      | 0.0           | 2196     | 1      | 37          | 2196      | 1          | 37    |
| NetBIOS Session Service                                             | 1.6             | 884     | 17.6          | 5485249  | 4,249  | 14          | 532       | 0          | 884   |
| Lightweight Directory Access Protocol                               | 2.3             | 1278    | 2.7           | 840142   | 650    | 1278        | 648868    | 502        | 1349  |
| Kerberos                                                            | 0.1             | 46      | 0.2           | 56566    | 43     | 46          | 56566     | 43         | 46    |
| Hypertext Transfer Protocol                                         | 0.0             | 3       | 0.0           | 859      | 0      | 2           | 241       | 0          | 4     |
| Distributed Computing Environment / Remote Procedure Call (DCE/RPC) | 3.0             | 1683    | 1.6           | 490128   | 379    | 473         | 223560    | 173        | 1683  |
| Data                                                                | 2.6             | 1414    | 6.3           | 1969980  | 1,526  | 1414        | 1969980   | 1,526      | 1414  |
| Internet Group Management Protocol                                  | 0.0             | 7       | 0.0           | 120      | 0      | 7           | 120       | 0          | 7     |
| Internet Control Message Protocol                                   | 0.0             | 4       | 0.0           | 160      | 0      | 4           | 160       | 0          | 4     |
| Address Resolution Protocol                                         | 3.7             | 2043    | 0.2           | 57204    | 44     | 2043        | 57204     | 44         | 2043  |

No display filter:

We can look at the http since they are little

And we see that asking for a dat file from an ip not a website

| No.  | Time        | Source          | Destination     | src port | d port | Protocol | Length | Info                                           |
|------|-------------|-----------------|-----------------|----------|--------|----------|--------|------------------------------------------------|
| 1721 | 106.503311  | 10.0.0.149      | 128.254.207.55  | 49750    | 80     | HTTP     | 141    | GET /86607.dat HTTP/1.1                        |
| 3656 | 108.530352  | 128.254.207.55  | 10.0.0.149      | 80       | 49750  | HTTP     | 965    | HTTP/1.1 200 OK                                |
| 8146 | 1395.163950 | 10.0.0.149      | 192.229.211.108 | 49896    | 80     | HTTP     | 208    | GET /DigiCertTLRSASHA2562020CA1-1.crt HTTP/1.1 |
| 8149 | 1395.206114 | 192.229.211.108 | 10.0.0.149      | 80       | 49896  | HTTP     | 150    | HTTP/1.1 200 OK (application/pkix-cert)        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| * Frame 1721: 141 bytes on wire (1128 bits), 141 bytes captured (1128 bits) on interface 0<br>* Ethernet II, Src: Intel_9e:42:fb (00:21:5d:9e:42:fb), Dst: Cisco_e5:77:18 (00:13:5f:e5:77:18)<br>* Internet Protocol Version 4, Src: 10.0.0.149, Dst: 128.254.207.55<br>* Transmission Control Protocol, Src Port: 49750, Dst Port: 80, Seq: 1, Ack: 1, Len: 87<br>* Hypertext Transfer Protocol<br>GET /86607.dat HTTP/1.1<br>Request Method: GET<br>Request URI: /86607.dat<br>Request Version: HTTP/1.1<br>Host: 128.254.207.55<br>User-Agent: curl/7.83.1<br>Accept: */*<br>[Response in frame: 3656]<br>[Full request URI: http://128.254.207.55/86607.dat] | 0000 00 13 5f e5 77 18 00 21 5d 9e 42 fb 08 00 45 00 ...w... ] B...E...<br>0010 00 7f e9 ca 40 00 80 06 b5 e3 0a 00 00 95 80 fe ...@... ..<br>0020 cf 37 c2 56 00 50 de 85 62 0a 0a 09 54 c9 50 18 ...7 V P... b...T P...<br>0030 fa f0 e4 73 00 00 47 45 54 20 2f 38 36 36 30 37 ...s...GE T /86607...<br>0040 2e 64 61 74 20 48 54 54 50 2f 31 2e 31 0d 0a 48 ...dat HTT P/1.1..H...<br>0050 6f 73 74 3a 20 31 32 38 2e 32 35 34 2e 32 30 37 ...ost: 128 .254.207...<br>0060 2e 35 35 0d 0a 55 73 65 72 2d 41 67 05 0e 74 3a ...55..Use r-Agent:<br>0070 20 63 75 72 6c 2f 37 2e 38 33 2e 31 0d 0a 41 63 ...curl/7. 83.1..Ac<br>0080 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0d 0a ...cept: */ *.... |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

We see that e have a program problem running  
MZ is digital signature ----> try to see on google search

```

GET /86607.dat HTTP/1.1
Host: 128.254.207.55
User-Agent: curl/7.83.1
Accept: */*

HTTP/1.1 200 OK
Server: nginx
Date: Fri, 03 Feb 2023 17:04:24 GMT
Content-Type: application/octet-stream
Content-Length: 1761280
Connection: keep-alive
Accept-Ranges: bytes
Expires: 0
Cache-Control: no-cache, no-store, must-revalidate
Content-Disposition: attachment;

MZ.....@.....!..L!This program cannot be run in DOS mode.
.....Rich.....P...d...@...s...!..
.....text.....rddata.s...
.....@...data...1...@...B.....sxdata...@...rsrc...@...@...
.....

```

Know get the file and see the hashes



713207d9d9875ec88d2f3a53377bf8c2d620147a4199eb183c13a7e957056432

55  
/72

Community Score

55/72 security vendors flagged this file as malicious

713207d9d9875ec88d2f3a53377bf8c2d620147a4199eb183c13a7e957056432

EslmgDet.dll

pedrlong-sleepscheck-user-inputdetect-debug-environmentpersistence

DETECTIONDETAILSRELATIONSBEHAVIORCOMMUNITY3

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat labeltrojan.qakbot/qakbotThreat categoriestrojan spywareFamily labelsqbotqakbotbebik

Security vendors' analysisDo you want to automate checks?

|             |                           |                  |                              |
|-------------|---------------------------|------------------|------------------------------|
| AhnLab-V3   | Trojan.Win.QakBot.R556576 | Alibaba          | Trojan.Win32.Qakbot.69ae4e02 |
| AliCloud    | Trojan.Win/Qbot.DZ        | ALYac            | Trojan.Agent.QakBot          |
| Antiy-AVL   | Trojan(Spy)/Win32.Bobik   | Arcabit          | Trojan.Generic.D400425A      |
| Arctic Wolf | Unsafe                    | Avast            | Win32:MalwareX-gen [Bot]     |
| AVG         | Win32:MalwareX-gen [Bot]  | Avira (no cloud) | TR/AD.KBot.elvuh             |

m8a8aBy@m8a8aBy-VirtualBox: ~/Desktop  
m8a8aBy@m8a8aBy-VirtualBox:~/Desktop\$ ls  
01\_Phishing\_Analysis 04\_SIEH 07\_Incident\_Response  
02\_Network\_Security 05\_Threat\_Intelligence 86607.dat  
03\_Endpoint\_Security 06\_Digital\_Forensics 500  
m8a8aBy@m8a8aBy-VirtualBox:~/Desktop\$ sha256sum 86607.dat  
Command 'sha256sum' not found, did you mean:  
command 'sha256sum' from deb coreutils (9.4-2ubuntu2)  
Try: sudo apt install <deb name>  
m8a8aBy@m8a8aBy-VirtualBox:~/Desktop\$ sha256sum 86607.dat  
713207d9d9875ec88d2f3a53377bf8c2d620147a4199eb183c13a7e957056432 86607.dat  
m8a8aBy@m8a8aBy-VirtualBox:~/Desktop\$

Try to look more about hash and see which family is it and what it can do and which protocols does it trigger



# Wireshark Challenge 1

Thursday, July 10, 2025

1:00 PM

## Challenge URL

- [https://challenges.malwarecube.com/  
#/c/7265ec1c-9773-4c7c-9ed4-2ea26e19f346](https://challenges.malwarecube.com/#/c/7265ec1c-9773-4c7c-9ed4-2ea26e19f346)



## Intrusion Detection and Prevention Systems (IDS/IPS)

- **Intrusion Detection Systems (IDS)**

- Passive monitoring and analysis
- Alert generation
- Logging
- Out-of-band

- **Intrusion Prevention Systems (IPS)**

- Traffic inspection
- Blocking and dropping packets
- Terminating connections
- Alert generation
- Logging
- Inline deployment

- **Network-Based IDS/IPS (NIDS/NIPS)**

- Operate at the network level
- Analyze traffic flowing through network appliances
- Alert, log, and/or block malicious traffic

- **Host-Based IDS/IPS (HIDS/HIPS)**

- Operate at the endpoint level
- Analyze activities on the host (file system, logins, apps)
- Alert, log, and/or block malicious host actions

## Detection Methods

- **Signature-based Detection**

- Comparing traffic or activities against known attack patterns
- Effective at identifying known threats
- Cannot detect what it doesn't know exists

- **Behavior-based Detection**

- Identifying anomalies that deviate from a pre-established baseline
- More effective at detecting unknown threats
- More likely to generate false positives

- **Rule-based Detection**

- Enforcing policies to define acceptable actions or violations
- Highly customizable to an organization's requirements
- Requires frequent maintenance and rule updates

|                  |                    |
|------------------|--------------------|
| Known<br>Known   | Known<br>Unknown   |
| Unknown<br>Known | Unknown<br>Unknown |

## Introduction to Snort

Thursday, July 10, 2025 12:57 PM

### What is Snort?

Snort is an open-source **network intrusion detection system (NIDS)** and **intrusion prevention system (IPS)** developed by Martin Roesch and maintained by Cisco.

#### Purpose:

It monitors network traffic in real time to **detect** and optionally **block** malicious activity.

#### Key Features:

- Packet sniffing and logging
- Attack detection using rules
- Real-time alerts
- Inline mode for blocking threats

#### Modes of Operation:

- **Sniffer Mode** – Displays traffic on the console
- **Packet Logger Mode** – Saves packets to disk
- **NIDS Mode** – Analyzes traffic and triggers alerts

#### Common Uses:

- Detecting attacks like port scans and buffer overflows
- Monitoring network traffic for anomalies
- Enhancing network security with real-time alerts

```
sudo apt install snort
```

Snort uses **Promiscuous mode**

is a network interface card (NIC) setting that allows the device to **intercept and read all traffic** on the network segment, not just the traffic addressed to it.

This is snort files having config files

```
m8a8a8y@m8a8a8y-VirtualBox:/$ cd /etc/snort
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ls -la
total 376
drwxr-xr-x  3 root root   4096 Jul 11 15:36 .
drwxr-xr-x 141 root root 12288 Jul 11 15:36 ..
-rw-r--r--  1 root root   1281 Apr 20  2022 attribute_table.dtd
-rw-r--r--  1 root root   3757 Apr 20  2022 classification.config
-rw-r--r--  1 root root  82469 Apr 19  2024 community-sid-msg.map
-rw-r--r--  1 root root  23654 Apr 20  2022 file_magic.conf
-rw-r--r--  1 root root  33339 Apr 20  2022 gen-msg.map
-rw-r--r--  1 root root    687 Apr 20  2022 reference.config
drwxr-xr-x  2 root root   4096 Jul 11 15:36 rules
-rw-r-----  1 root snort 29773 Apr 19  2024 snort.conf
-rw-----  1 root root    803 Jul 11 15:36 snort.debian.conf
-rw-r--r--  1 root root   2335 Apr 20  2022 threshold.conf
-rw-r--r--  1 root root 160606 Apr 20  2022 unicode.map
```

The best thing is to make backup from snort.conf

```
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo cp snort.conf snort.conf.bak
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ls
attribute_table.dtd  community-sid-msg.map  gen-msg.map          rules                snort.conf.bak       threshold.conf
classification.config  file_magic.conf       reference.config     snort.conf          snort.debian.conf    unicode.map
```

Now open the configuration file

```
sudo nano snort.conf
```

```
# This file contains a sample snort configuration.
# You should take the following steps to create your own custom configuration:
#
# 1) Set the network variables.
# 2) Configure the decoder
# 3) Configure the base detection engine
# 4) Configure dynamic loaded libraries
# 5) Configure preprocessors
# 6) Configure output plugins
# 7) Customize your rule set
# 8) Customize preprocessor and decoder rule set
# 9) Customize shared object rule set
```

And can be changed

```
#####
# Step #1: Set the network variables. For more information, see README.variables
#####

# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET any
```

← Done on installation

This path is used to make your own custom rules

```
#####
# Step #7: Customize your rule set
# For more information, see Snort Manual, Writing Snort Rules
#
# NOTE: All categories are enabled in this conf file
#####

# Note to Debian users: The rules preinstalled in the system
# can be *very* out of date. For more information please read
# the /usr/share/doc/snort-rules-default/README.Debian file
#
# If you install the official VRT Sourcefire rules please review this
# configuration file and re-enable (remove the comment in the first line) those
# rules files that are available in your system (in the /etc/snort/rules
# directory)

# site specific rules
include $RULE_PATH/local.rules
```

To make your own rules work make all rules in step 7 commented by adding ( # ) before each line

then save and test if there is no problems

```
sudo snort -T -c /etc/snort/snort.conf
```

```
+++++
Initializing rule chains...
0 Snort rules read
  0 detection rules
  0 decoder rules
  0 preprocessor rules
0 Option Chains linked into 0 Chain Headers
+++++

+-----[Rule Port Counts]-----+
|      tcp      udp      icmp      ip
|  src          0          0          0
|  dst          0          0          0
|  any          0          0          0
|  nc           0          0          0
|  s+d          0          0          0
+-----+

```

```

Using ZLIB version: 1.3

Rules Engine: SF_SNORT_DETECTION_ENGINE Version 3.2 <Build 1>
Preprocessor Object: SF_DNP3 Version 1.1 <Build 1>
Preprocessor Object: appid Version 1.1 <Build 5>
Preprocessor Object: SF_SSH Version 1.1 <Build 3>
Preprocessor Object: SF_IMAP Version 1.0 <Build 1>
Preprocessor Object: SF_SIP Version 1.1 <Build 1>
Preprocessor Object: SF_FTPTELNET Version 1.2 <Build 13>
Preprocessor Object: SF_POP Version 1.0 <Build 1>
Preprocessor Object: SF_DNS Version 1.1 <Build 4>
Preprocessor Object: SF_REPUTATION Version 1.1 <Build 1>
Preprocessor Object: SF_SDF Version 1.1 <Build 1>
Preprocessor Object: SF_SSLPP Version 1.1 <Build 4>
Preprocessor Object: SF_S7COMMPLUS Version 1.0 <Build 1>
Preprocessor Object: SF_MODBUS Version 1.1 <Build 1>
Preprocessor Object: SF_SMTP Version 1.1 <Build 9>
Preprocessor Object: SF_GTP Version 1.1 <Build 1>
Preprocessor Object: SF_DCCRPC2 Version 1.0 <Build 3>

Total snort Fixed Memory Cost - MaxRss:48044
Snort successfully validated the configuration!
snort exiting

```

## Snort modes

### 1. Sniffer mode

```

n8a8a8y@m8a8a8y-VirtualBox:~$ sudo snort -i enp0s3
[sudo] password for n8a8a8y:
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

--== Initialization Complete ==--

o^_)-  -> Snort! <*-
  '---'  Version 2.9.20 GRE (Build 82)
          By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
          Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
          Copyright (C) 1998-2013 Sourcefire, Inc., et al.
          Using libpcap version 1.10.4 (with TPACKET_V3)
          Using PCRE version: 8.39 2016-06-14
          Using ZLIB version: 1.3

Commencing packet processing (pid=4023)
07/11-16:10:26.308329 10.0.2.15:54339 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:44925 IpLen:20 DgmLen:67
Len: 39
=====

07/11-16:10:26.308981 10.0.2.15:55396 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:27052 IpLen:20 DgmLen:67
Len: 39
=====

WARNING: No preprocessors configured for policy 0.
07/11-16:10:26.327380 10.0.2.3:53 -> 10.0.2.15:55396
UDP TTL:64 TOS:0x0 ID:3464 IpLen:20 DgmLen:95
Len: 67

```

```

n8a8a8y@m8a8a8y-VirtualBox:~$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
n8a8a8y@m8a8a8y-VirtualBox:~$

```

### Adding -e will give more info

```

n8a8a8y@m8a8a8y-VirtualBox:~$ sudo snort -i enp0s3 -e
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

--== Initialization Complete ==--

o^_)-  -> Snort! <*-
  '---'  Version 2.9.20 GRE (Build 82)
          By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
          Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
          Copyright (C) 1998-2013 Sourcefire, Inc., et al.
          Using libpcap version 1.10.4 (with TPACKET_V3)
          Using PCRE version: 8.39 2016-06-14
          Using ZLIB version: 1.3

Commencing packet processing (pid=4059)
07/11-16:14:26.525081 08:00:27:A4:86:ED -> 52:55:0A:00:02:02 type:0x800 len:0x4A
10.0.2.15:48280 -> 142.251.37.46:80 TCP TTL:64 TOS:0x0 ID:53627 IpLen:20 DgmLen:60 DF
*****S* Seq: 0xD6C78E0A Ack: 0x0 Win: 0xFAF0 TcpLen: 40
TCP Options (5) => MSS: 1460 SackOK TS: 3191672462 0 NOP WS: 7
=====

WARNING: No preprocessors configured for policy 0.
07/11-16:14:26.585807 52:55:0A:00:02:02 -> 08:00:27:A4:86:ED type:0x800 len:0x3C
142.251.37.46:80 -> 10.0.2.15:48280 TCP TTL:64 TOS:0x0 ID:3489 IpLen:20 DgmLen:44
***A**S* Seq: 0xA4EAC01 Ack: 0xD6C78E0B Wln: 0xFFFF TcpLen: 24
TCP Options (1) => MSS: 1460
=====

```



Adding the -d will give you the ASCII

```

mash@kali:~$ sudo snort -i enp0s3 -d
Running in packet dump mode

== Initializing Snort ==
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

== Initialization Complete ==

--> Snort! <*-
''- Version 2.9.20 GRE (Build 82)
**** By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
      Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
      Copyright (C) 1998-2013 Sourcefire, Inc., et al.
      Using libpcap version 1.10.4 (with PACKET_V3)
      Using PCRE version: 8.39 2016-06-14
      Using ZLIB version: 1.3

Commencing packet processing (pid=4081)
07/11-16:17:26.086931 10.0.2.15:57363 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:7656 ILen:20 DgLen:67
Len: 39
17 DC 01 00 00 01 00 00 00 00 01 06 67 6F 6F .....goo
67 6C 65 03 63 6F 6D 00 00 1C 00 01 00 29 05 gle.com.....).
C0 00 00 00 00 00 00 .....
=====

WARNING: No preprocessors configured for policy 0.
07/11-16:17:26.105101 10.0.2.3:53 -> 10.0.2.15:57363
UDP TTL:64 TOS:0x0 ID:3500 ILen:20 DgLen:95
Len: 67
17 DC 81 80 00 01 00 01 00 00 00 01 06 67 6F 6F .....goo

```

## 2. Packet logger mode

```

m8asasy@m8asasy-VirtualBox:~/logs$ sudo snort -i enp0s3 -l .
Running in packet logging mode

--== Initializing Snort ==--
Initializing Output Plugins!
Log directory = .
pcap DAQ configured to passive.
Acquiring network traffic from "enp0s3".
Decoding Ethernet

--== Initialization Complete ==--

**=
o*)-  .*= Snort! <*=
****  Version 2.9.20 GRE (Build 82)
      By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
      Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
      Copyright (C) 1998-2013 Sourcefire, Inc., et al.
      Using libpcap version 1.10.4 (with TPACKET_V3)
      Using PCRE version: 8.39 2016-06-14
      Using ZLIB version: 1.3

Commencing packet processing (pid=4152)
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
WARNING: No preprocessors configured for policy 0.
^C*** Caught Int-Signal
WARNING: No preprocessors configured for policy 0.
=====

```

```
m8sa8aSy@m8sa8aSy-VirtualBox:~/logs$ ls
snort.log.1752240294
m8sa8aSy@m8sa8aSy-VirtualBox:~/logs$ sudo snort -r snort.log.1752240294
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1752240294".

--== Initialization Complete ==--

^C
o")~      -*> Snort! *-
****      Version 2.9.20 GRE (Build 82)
          By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
          Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
          Copyright (C) 1998-2013 Sourcefire, Inc., et al.
          Using libcap version 1.10.4 (with TPACKET_V3)
          Using PCRE version: 8.39 2016-06-14
          Using ZLIB version: 1.3

Commencing packet processing (pid=4175)
07/11-16:24:56.527763 10.0.2.15:60468 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:25725 IpLen:20 DgmLen:67
Len: 39
*****
```

```

m8a8a8y@m8a8a8y-VirtualBox:~/logs$ ls
snort.log.1752240294
m8a8a8y@m8a8a8y-VirtualBox:~/logs$ sudo snort -r snort.log.1752240294
Running in packet dump mode

--== Initializing Snort ==--
Initializing Output Plugins!
pcap DAQ configured to read-file.
Acquiring network traffic from "snort.log.1752240294".

--== Initialization Complete ==--

o''-~
''''
-*)<*-
Version 2.9.20 GRE (Build 82)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.4 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.3

Commencing packet processing (pid=4175)
07/11-16:24:56.527763 10.0.2.15:60468 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:25725 IpLen:20 DgmLen:67
Len: 39
=====
07/11-16:24:56.528721 10.0.2.15:52357 -> 10.0.2.3:53
UDP TTL:64 TOS:0x0 ID:4522 IpLen:20 DgmLen:67
Len: 39
=====

```

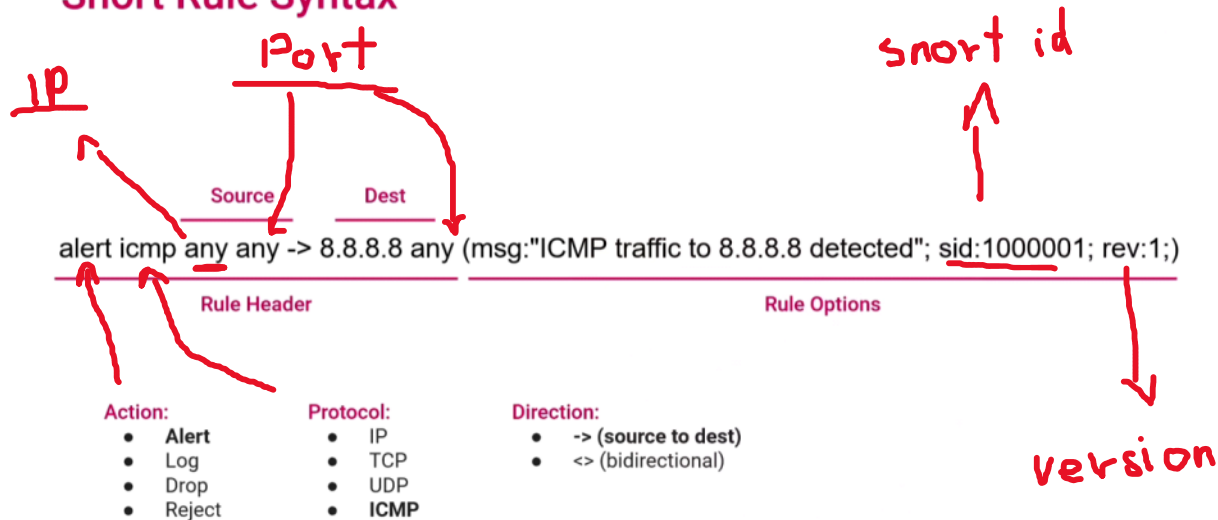
Also tcpdump can use this file

```

m8a8a8y@m8a8a8y-VirtualBox:~/logs$ sudo tcpdump -r snort.log.1752240294
reading from file snort.log.1752240294, link-type EN10MB (Ethernet), snapshot length 1514
16:24:56.527763 IP m8a8a8y-VirtualBox.60468 > 10.0.2.3.domain: 38191+ [1au] A? google.com. (39)
16:24:56.528721 IP m8a8a8y-VirtualBox.52357 > 10.0.2.3.domain: 13458+ [1au] AAAA? google.com. (39)
16:24:56.543472 IP 10.0.2.3.domain > m8a8a8y-VirtualBox.52357: 13458 1/0/1 AAAA 2a00:1450:4006:810::200e (67)
16:24:56.543477 IP 10.0.2.3.domain > m8a8a8y-VirtualBox.60468: 38191 1/0/1 A 142.251.37.46 (55)
16:24:56.545836 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [S], seq 331796290, win 64240, options [mss 1460,sackOK,TS val 3192302482 ecr 0,nop,wscale 7], length 0
16:24:56.610211 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [S.], seq 182016001, ack 1331796291, win 65535, options [mss 1460], length 0
16:24:56.610336 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [.], ack 182016001, win 64240, length 0
16:24:56.610873 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [P.], seq 1:74, ack 1, win 64240, length 73: HTTP: GET / HTTP/1.1
16:24:56.611646 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [.], ack 74, win 65535, length 0
16:24:56.764251 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [P.], seq 1:774, ack 74, win 65535, length 773: HTTP: HTTP/1.1 301 Moved Permanently
16:24:56.764409 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [.], ack 774, win 63467, length 0
16:24:56.765001 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [F.], seq 774, ack 774, win 63467, length 0
16:24:56.765328 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [.], ack 774, win 65535, length 0
16:24:56.827739 IP mrs09s13-in-f14.1e100.net.http > m8a8a8y-VirtualBox.60556: Flags [F.], seq 774, ack 75, win 65535, length 0
16:24:56.827771 IP m8a8a8y-VirtualBox.60556 > mrs09s13-in-f14.1e100.net.http: Flags [.], ack 75, win 63467, length 0
16:25:02.000548 ARP, Request who-has _gateway tell m8a8a8y-VirtualBox, length 28

```

## Snort Rule Syntax



```
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ls
attribute_table.dtd  community-sid-msg.map  gen-msg.map  rules  snort.conf.bak  threshold.conf
classification.config  file_magic.conf  reference.config  snort.conf  snort.debian.conf  unicode.map
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo nano /rules/local.rules
```

```
GNU nano 7.2 local.rules
## $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
```



```

GNU nano 7.2                               rules/local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> 8.8.8.8 any (msg: "ICMP traffic to 8.8.8.8 detected"; sid: 1000001; rev:1;)
```

-A alert-mode

Alert using the specified **alert-mode**. Valid alert modes include **fast**, **full**, **none**, and **unsock**. **Fast** writes alerts to the default "alert" file in a single-line, syslog style alert message. **Full** writes the alert to the "alert" file with the full decoded header as well as the alert message. **None** turns off alerting. **Unsock** is an experimental mode that sends the alert information out over a UNIX socket to another process that attaches to that socket.

```
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo snort -A console -l /var/log/snort -i enp0s3 -c /etc/snort/snort.conf -q
```

Display alerts on the command

interface

Config file of snort

Quite mode no details  
If not used its OK

Saving log

If we ping 8.8.4.4

```

m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo snort -A console -l /var/log/snort/ -i enp0s3 -c /etc/snort/snort.conf -q
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ping 8.8.4.4
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.
64 bytes from 8.8.4.4: icmp_seq=1 ttl=255 time=65.2 ms
64 bytes from 8.8.4.4: icmp_seq=2 ttl=255 time=66.4 ms
64 bytes from 8.8.4.4: icmp_seq=3 ttl=255 time=69.5 ms
64 bytes from 8.8.4.4: icmp_seq=4 ttl=255 time=65.4 ms
64 bytes from 8.8.4.4: icmp_seq=5 ttl=255 time=66.8 ms
^C
--- 8.8.4.4 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4009ms
rtt min/avg/max/mdev = 65.236/66.659/69.466/1.520 ms
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$
```

And if 8.8.8.8

```

m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ sudo snort -A console -l /var/log/snort/ -i enp0s3 -c /etc/snort/snort.conf -q
07/12-11:25:10.015456 1:1000001:1 ICMP from 8.8.8.8 altered [**] [Priority: 0] {ICMP}
10.0.2.15 -> 8.8.8.8
07/12-11:25:11.017576 1:1000001:1 ICMP from 8.8.8.8 altered [**] [Priority: 0] {ICMP}
10.0.2.15 -> 8.8.8.8
07/12-11:25:12.018733 1:1000001:1 ICMP from 8.8.8.8 altered [**] [Priority: 0] {ICMP}
10.0.2.15 -> 8.8.8.8
07/12-11:25:13.021554 1:1000001:1 ICMP from 8.8.8.8 altered [**] [Priority: 0] {ICMP}
10.0.2.15 -> 8.8.8.8
07/12-11:25:14.023590 1:1000001:1 ICMP from 8.8.8.8 altered [**] [Priority: 0] {ICMP}
10.0.2.15 -> 8.8.8.8
07/12-11:25:15.024953 1:1000001:1 ICMP from 8.8.8.8 altered [**] [Priority: 0] {ICMP}
10.0.2.15 -> 8.8.8.8
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=67.9 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=65.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=70.5 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=65.1 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=65.4 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=66.8 ms
^C
--- 8.8.8.8 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 65.138/66.854/70.455/1.876 ms
m8a8a8y@m8a8a8y-VirtualBox:/etc/snort$
```

<https://snort.org/downloads#rule-downloads> : has many ready rules can be used

```
tcm@S0C101-ubuntu:/var/log/snort$ ls
snort.alert.fast  snort.log.1717962723  snort.log.1717963124
tcm@S0C101-ubuntu:/var/log/snort$ sudo wireshark
```

| Apply a display filter: *Ctrl+L* |                            |             |             |               |           |          |        |             |                           |
|----------------------------------|----------------------------|-------------|-------------|---------------|-----------|----------|--------|-------------|---------------------------|
| No.                              | Time                       | Source      | Source Port | Destination   | Dest Port | Protocol | Length | Info        |                           |
| 1                                | 2024-06-09 19:58:54.422137 | 192.168.1.4 | 2125        | 93.184.215.14 | 4444      | TCP      | 54     | 2125 → 4444 | [SYN] Seq=0 Win=512 Len=0 |

U can use snorpy -> <https://snorpy.cyb3rs3c.net/>  
To make rules

## SNORPY

A Web Based Snort Rule Creator / Maker for Building Simple Snort Rules

|                                             |     |     |     |   |            |          |         |   |
|---------------------------------------------|-----|-----|-----|---|------------|----------|---------|---|
| alert                                       | tcp | any | any | » | 8.8.8.8    | 4444     | 1000001 | 1 |
| Rule Message ( \ Escape special characters) |     |     |     |   | Class-Type | Priority | gid     |   |

### TCP

HTTP REQUEST METHOD  HTTP RESPONSE CODE

ACK ☐ SYN ☐ PSH ☐ RST ☐ FIN ☐ URG ☐ \* ☐ \*

DIRECTION  TCP STATE

Data Size

Reference

Threshold Tracking Type  TRK BY  Count #  Seconds

Add Content Match

Add Regex Match

alert tcp any any -> 8.8.8.8 4444 ( skd:1000001; rev:1; )

Network Security Page 34

# Snort: Intrusion Detection and Prevention

Thursday, July 10, 2025 12:57 PM

We are going to make an alert in exe file on a pcap

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo wireshark 1.pcap
[sudo] password for m8a8a8y:
** (wireshark:2538) 20:33:17.384310 [GUI WARNING] -- QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
```

SNORPY

A Web Based Snort Rule Creator / Maker for Building Simple Snort Rules

alert tcp any any -> any 80 ( msg:'HTTP contains .exe'; content:'|2e|exe'; nocase; http\_uri; sid:10000001; rev:1; )

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo nano /etc/snort/rules/local.rules
```

```
GNU nano 7.2 /etc/snort/rules/local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert tcp any any -> any 80 ( msg:'HTTP contains .exe'; content:'|2e|exe'; nocase; http_uri; sid:10000001; rev:1; )
```

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo snort -c /etc/snort/snort.conf -q -r 1.pcap -A console
09/14-17:35:31.840413 [**] [1:10000001:1] HTTP contains .exe [**] [Priority: 0] {TCP} 10.0.0.168:49722 -> 13.107.5.80:80
09/14-17:35:32.552578 [**] [1:10000001:1] HTTP contains .exe [**] [Priority: 0] {TCP} 10.0.0.168:49724 -> 103.232.55.148:80
```

```
m8a8a8y@m8a8a8y-VirtualBox:~/Desktop/02_Network_Security/03_Snort/PCAPs$ sudo snort -q -d -r /var/log/snort/snort.log.1752342302
09/14-17:35:31.840413 10.0.0.168:49722 -> 13.107.5.80:80
TCP TTL:128 TOS:0x0 ID:502 IpLen:20 DgmLen:706 DF
***AP*** Seq: 0x53A34ABB Ack: 0xD074CC7D Win: 0x7FAE TcpLen: 20
47 45 54 20 2F 71 73 6D 6C 2E 61 73 70 78 3F 71 GET /qsm1.aspx?q
75 65 72 79 3D 68 74 74 70 25 33 41 25 32 46 25 uery=http%3A%2F%
32 46 31 30 33 2E 32 33 32 2E 35 35 2E 31 34 38 2F103.232.55.148
25 32 46 73 65 72 76 69 63 65 25 32 46 2E 61 75 %2Fservice%2F.au
64 69 6F 64 67 2F 65 78 65 26 6D 61 78 77 69 64 diodn.exe&maxwid
```

*Handwritten red notes:* "decode" with an arrow pointing to the packet details, and "suite" with an arrow pointing to the packet details.

```

47 45 54 20 2F 71 73 6D 6C 2E 61 73 70 78 3F 71 GET /qsm.aspx?q
75 65 72 79 3D 68 74 74 70 25 33 41 25 32 46 25 uery=http%3A%2F%
32 46 31 30 33 2E 32 33 32 2E 35 35 2E 31 34 38 2F103.232.55.148
25 32 46 73 65 72 76 69 63 65 25 32 46 2E 61 75 %2Fservice%2F.au
64 69 6F 64 67 2E 65 78 65 26 6D 61 78 77 69 64 diodg.exe&maxwid
74 68 3D 33 32 37 36 35 26 72 6F 77 68 65 69 67 th=32765&rowheig
68 74 3D 32 30 26 73 65 63 74 69 6F 6E 48 65 69 ht=20&sectionHei
67 68 74 3D 31 36 30 26 46 4F 52 4D 3D 49 45 53 ght=160&FORM=IES
53 30 32 26 6D 61 72 68 65 74 3D 65 6E 2D 55 53 S02&market=en-US
20 48 54 54 50 2F 31 2E 31 0D 0A 41 63 63 65 70 HTTP/1.1..Accep
74 3A 20 2A 2F 2A 0D 0A 41 63 63 65 70 74 2D 4C t: /*..Accept-L
61 6E 67 75 61 67 65 3A 20 65 6E 2D 55 53 0D 0A nguage: en-US..
55 73 65 72 2D 41 67 65 6E 74 3A 20 4D 6F 7A 69 User-Agent: Mozi
6C 6C 61 2F 35 2E 30 28 57 69 6E 64 6F 77 73 lla/5.0 (Windows
20 4E 54 20 31 30 2E 30 38 20 57 4F 57 36 34 38 NT 10.0; WOW64;
20 54 72 69 64 65 6E 74 2F 37 2E 30 38 20 72 76 Trident/7.0; rv
3A 31 31 2E 30 29 20 6C 69 68 65 20 47 65 63 68 :11.0) like Geck
6F 0D 0A 41 63 63 65 70 74 2D 45 6E 63 6F 64 69 o..Accept-Encodi
6E 67 3A 20 67 7A 69 70 2C 20 64 65 66 6C 61 74 ng: gzip, deflat
65 0D 0A 48 6F 73 74 3A 20 61 70 69 2E 62 69 6E e..Host: api.bin
67 2E 63 6F 6D 0D 0A 43 6F 6E 6E 65 63 74 69 6F g.com..Connectio
6E 3A 20 48 65 65 70 2D 41 6C 69 76 65 0D 0A 43 n: Keep-Alive..C
6F 6F 68 69 65 3A 20 4D 55 49 44 3D 33 42 31 32 ookie: MUID=3B12
36 41 38 46 35 38 41 45 36 45 44 37 33 39 38 41 6A8F58AE6ED7398A
37 41 33 44 35 39 30 44 36 46 34 35 38 20 5F 45 7A3D590D6F45; _E
44 47 45 5F 53 3D 46 3D 31 26 53 49 44 3D 33 38 DGE S=F=1&SID=38

```

quite

Not files are being sent with .exe some are transmitted as  
And we can use it to trigger alert

Wireshark - Follow HTTP Stream (tcp.stream eq 40) - 1.pcap

GET /service/audiogd.exe HTTP/1.1  
Accept: text/html, application/xhtml+xml, image/jxr, /\*/  
Accept-Language: en-US  
User-Agent: Mozilla/5.0 (Windows NT 10.0; WOW64; Trident/7.0; rv:11.0) like Gecko  
Accept-Encoding: gzip, deflate  
Host: 103.232.55.148  
Connection: Keep-Alive

HTTP/1.1 200 OK  
Date: Tue, 14 Sep 2021 14:35:37 GMT  
Server: Apache/2.4.47 (win64) OpenSSL/1.1.1k PHP/7.3.28  
Last-Modified: Mon, 13 Sep 2021 20:13:53 GMT  
ETag: "92a00-5cbe61aac8772"  
Accept-Ranges: bytes  
Content-Length: 600576  
Keep-Alive: timeout=5, max=100  
Connection: Keep-Alive  
Content-Type: application/x-msdownload

MZ:.....@.....!..L!This program cannot be run in DOS mode.

And here we want the response header on port 80

# SNORPY

A Web Based Snort Rule Creator / Maker for Building Simple Snort Rules

alert ▾ tcp ▾ any 80 » any any 10000001 ?

HTTP contains .exe
Class-Type
Priority ▾ glid

### TCP

HTTP REQUEST METHOD ▾
HTTP RESPONSE CODE ▾

ACK ☐ SYN ☐ PSH ☐ RST ☐ FIN ☐ URG ☐ + ☐ \*

DIRECTION ▾
TCP STATE ▾

Data Size ▾

Reference ▾

Threshold Tracking Type ▾
TRK BY ▾
Count #
Seconds

Content-Type: application/x-msdownload

Offset
Depth
nocase ☒ uri ☒ not ☐

✓
↔
✗

Add Regex Match  
+

```

alert tcp any 80 -> any any ( msg:"HTTP contains .exe"; content:"[2e]exe"; nocase; http_uri; content:"Content[2d]Type[3a
20]application[2f]x[2d]msdownload"; nocase; http_uri; sid:10000001; rev:1; )

```

Try to make the rest alone

Aslo we can use the files siganture



# Snort Challenge 1

Thursday, July 10, 2025

12:57 PM

<https://challenges.malwarecube.com/#/c/fec27521-e154-4637-a1bd-38777d85c974>



# Additional Practice

Thursday, July 10, 2025 12:58 PM

## References

- <https://www.malware-traffic-analysis.net/>
- <https://chrissanders.org/packet-captures/>
- <https://github.com/chrissanders/packets>
- <https://wiki.wireshark.org/SampleCaptures>
- <https://www.netresec.com/?page=PcapFiles>
- <https://github.com/zeek/zeek/tree/master/testing/btest/Traces>
- <https://docs.securityonion.net/en/2.4/pcaps.html>